



An Coimisiún
um Rialáil Fónas
**Commission for
Regulation of Utilities**

An Coimisiún um Rialáil Fónas
Commission for Regulation of Utilities

Uisce Éireann Performance Assessment Framework

2021 Annual Report

Information Paper

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CRU Strategic Plan 2022-24

Our Mission • Protecting the public interest in water, energy and energy safety.	Our Strategic Priorities • Ensure Security of Supply • Drive a Low Carbon Future • Empower and Protect Customers • Enable our People and Organisational Capacity
Our Vision • Safe, secure and sustainable supplies of energy and water, for the benefit of customer now and in the future	

Executive Summary

The CRU is the regulator of Uisce Éireann as the national utility for the provision of public water and wastewater services. The CRU's role is to protect the interests of water and wastewater customers, ensure the delivery of water services in a safe, secure and sustainable manner and ensure that Uisce Éireann operates in an economic and efficient manner.

The CRU is responsible for setting the level of revenue that Uisce Éireann can receive, through various charges approved under the Water Charges Plan (new connections, non-domestic tariffs, etc) and also via the Government subvention. The CRU does this by reviewing Uisce Éireann's submissions, engaging with the utility, benchmarking its proposed costs against comparator companies, completing a public consultation process and setting appropriate revenue allowances for operational costs, capital costs and other items. This process is known as a revenue control. The current revenue control, Revenue Control 3 (RC3) runs from the period 2020 to 2024.

In August 2020, the Commission for Regulation of Utilities, (CRU) approved Uisce Éireann's Capital Investment Plan for the years 2020 – 2024 inclusive¹ under the CRU's Revenue Control 3 (RC3) process and decision. This decision set out the total revenue allowance approved by the CRU, to enable Uisce Éireann to efficiently meet and deliver its capital investment commitments and operational expenditure requirements across the five-year period. The RC3 decision followed a detailed assessment and engagement of submissions made by Uisce Éireann of its business plans for the five-year period.

Under RC3, the CRU set an allowed revenue for capital and operating expenditure of over €8 billion for Uisce Éireann. It is therefore vital that Uisce Éireann can demonstrate progress made in delivering its investment and service delivery targets.

As part of the RC3 process, the CRU monitors and publishes Uisce Éireann's performance against its Capital Investment Plan, Performance Assessment Framework and Customer Handbook. This information paper is the CRU's annual monitoring report of Uisce Éireann's progress in delivering against its **Performance Assessment Framework** under RC3. The Framework is designed to assess Uisce Éireann's overall performance in delivering defined services to its customers against the targets set by the CRU. This is the CRU's second assessment under the revised Performance Assessment Framework for 2020-2024. Uisce

¹ [CRU20085 Update to Uisce Éireann's Revenue Control 3 \(RC3.5\) – Uisce Éireann's Updated Capital Investment Plan](#)

Éireann's report under this revised Framework regarding its 2021 performance is published alongside this report (CRU202340a).

The updated Performance Assessment Framework requires Uisce Éireann to report on 27 metrics under six categories:

- Customer service
- Security of water supply,
- Quality of water supply,
- Sewer incidents,
- Environmental performance and
- Energy and emissions.

The metrics that Uisce Éireann is not yet reporting on in full and require further establishment before they are reported on are:

- Revised Customer Satisfaction Survey, Stakeholder Engagement Panel (CRU to establish), Security of Supply, Customer Side Pipe Leakage, Interruptions to supply – Minutes of Lost Supply, Wastewater Sludge Reuse and Disposal, Internal Sewer Incidents and External Sewer Incidents

The CRU set 2021 targets for Uisce Éireann under 12 of the 27 metrics. Some of these metrics have multiple targets. A brief assessment of Uisce Éireann's 2021 performance under each metric with a 2021 target is provided in summary table 1 below.

In 2021, Uisce Éireann met nine and missed ten of the 19 targets set for it that year. While some areas have observed improvements and targets have been met, there are metrics where targets have not been met and the gap that remains against targets set is of concern.

The CRU notes that, as the final performance targets under the Framework were confirmed in Q3 2021, Uisce Éireann was not in a position to sufficiently review or improve its processes to meet all the 2020 or 2021 targets, where required.

However, it is important to note that, these targets were based on the allowed spend over RC3(€8 billion over the RC 3 period); previous Uisce Éireann performance; and the performance of comparator utilities in a similar timeframe. The CRU would therefore expect Uisce Éireann to meet or get close to the targets set.

Table 1 Performance Against Targets, sets out Uisce Éireann's progress against the agreed targets under the performance assessment framework.

Summary Table 1. Performance Against Targets

Metric Name		2020 PAF Score	2021 PAF Score	2024 PAF Target	Assess	
Customer Service	Ease of telephone contact: Speed of telephone response		82%	76%	85%	✗
	Ease of telephone contact: Call abandonment rate		3%	4.40%	4%	✓
	Ease of telephone contact: First call resolution		86% Billing 75% Operations	77% Billing 71% Operations	90% Billing 80% Operations	✗
	Billing of metered customers	% bills issued based on a meter read	69%	78%	70%	✓
		% metered accounts receiving at least one bill based on a meter read	IW to set up metric	85%	100%	✗
	Response to billing contacts		96%	95.00%	95%	✓
	Response to complaints	% complaints responded to within 5 working days, either with a resolution or an outline plan of the proposed resolution	Domestic: 98% Non-domestic: 94%	Domestic: 94% Non-domestic: 90%	100%	✗
		percentage of complaints to which a final decision is issued within 2 months	Domestic: 94% Non-domestic: 89%	Domestic: 96% Non-domestic: 88%	100%	✗
	Unresolved complaints upheld by the CRU CCT		0.02 complaints upheld per 10,000 customers/sites	0.04 complaints upheld per 10,000 customers/sites.	Better than average of utilities or suppliers monitored by CRU CCT	✓
	Customer Satisfaction Survey		77%	78.50%	Establish baseline under new survey	CRU to engage with UÉ in 2022 on new survey
	Stakeholder Engagement					

Security of Water Supply	Security of Supply	Security of Supply Index		N/A	National Water Resources Plan Framework published June 2021	Publish National Water Resources Plan Framework and 4 Regional Plans by end 2021	✗
		number of water resource zones in deficit and the population served by those resource zones		N/A	No Regional Plans published by end 2021		
	Leakage	2019 baselines		N/A	Not submitted	IW to submit 2019 baselines by 31st December 2021	✗
		the amount of water lost on the public network		38 MI/day reduction	79 MI/day reduction	161 MI/day reduction	✓
		the amount of water lost on customer supply pipes		Not yet reported	Not yet reported	15 MI/day reduction	CRU to engage with UÉ in 2023 on baseline and methodology
	Quality of Water Supply	Interruptions to Supply	Minutes of lost supply		N/A	IW submitted plan to report on this metric in Q1 2022	Establish baseline performance
unplanned interruption			>4 hours	63.56%	65.90%	Trend downwards	✓
			>12 hours	20.36%	25.30%	<12% connected properties	✗
			>24 hours	7.61%	6.40%	<3.6% connected properties	✓
Drinking Water Quality		Microbiological compliance	99.96%	99.96%	99.9% min.	✓	
		E. coli compliance	99.96%	99.95%	99.9% min.	✓	

		Enterococci compliance	99.91%	100%	99.9% min.	✓
		Chemical compliance	99.71%	99.60%	99.9% min.	✓
		THM compliance	97.34%	95.30%	99.0% min.	✗
		Lead compliance	98.61%	98.92%	98.48% min.	✓
	Boil Water Notices and Drinking Water Restriction Notices	Number of public supplies on BWNs > 30 days	26	29	0	✗
		Population served by supplies on BWNs > 30 days	15,493	46,069	0	✗
		The number of public supplies on DWRNs > 30 days	8	9	0	✗
		Population served by supplies on DWRNs > 30 days	952	1,208	0	✗
Sewer Incidents	Internal Sewer Incidents (Overload)		IW to first report in 2022 (2021 data)	Not reported	To be set when baseline performance established in 2022	✗
	Internal Sewer Incidents (Other Causes)		IW to first report in 2022 (2021 data)	Not reported	To be set when baseline performance established in 2022	✗
	Internal Sewer Incidents (Properties at Risk)		IW to first report in 2022 (2021 data)	Not reported	To be set after engagement on target in 2022	✗
	External Sewer Incidents (Overload)		Not reported	Not reported	Establish baseline performance	✗
	External Sewer Incidents (Other Causes)		Not reported	Not reported	Establish baseline performance	✗

	External Sewer Incidents (Properties at Risk)		Not reported	Not reported	Establish baseline performance	✗
Environmental Performance	Incidents Relating to Wastewater	Once-off incidents	1,108	1,067	<345	✗
		Recurring incidents	238	249	<98	✗
		Category 3-5 incidents	1	0	0	✓
	Wastewater Agglomerations Meeting Treatment Requirements: Agglomerations with no Wastewater Treatment		34	33	0	✗
	Compliance with the Emission Limit Values for Urban Wastewater Licences	Overall compliance	41%	43%	Establish baseline performance	✓
		BOD limit	77%	77%		
		COD limit	90%	86%		
		Suspended solids limit	79%	77%		
		Ortho Phosphate limit	65%	70%		
		Ammonia limit	56%	60%		
Compliance with the treatment requirements of Urban Waste Water Treatment Directive		93%	93%	100%	✗	
Sludge Reuse and Disposal	Drinking water sludge	100%	100%	100%	✓	
	Wastewater sludge	Reporting capability to be in place Q4 2021	Reporting capability in place Q4 2021, reporting 2023 (2022 data)	First reporting 2023 (2022 data)	✓	
Energy and Emissions	Energy consumption		1,059.6 GWh (TPER)	TBC	>40.71 GWh reduction (TPER)	✓
	GHG emissions		TBC	TBC	TBD when scope of emissions target decided by the SEAI	N/A
KEY: ✓ - on track ✗ - not on track						

The CRU has identified some trends in Uisce Éireann's performance based on data submitted for 2020 and 2021. It noted that trends are not yet fully evident across the board on all metrics, as some metrics that reported acceptable or good performance in 2020, reported drops in performance in 2021 and vice versa. The CRU considers that Uisce Éireann's annual performance is key for customers and is important for the utility to address when implementing performance improvements.

The monitoring assessment and report assists the CRU to understand:

- Key areas of improved or maintained performance in 2021
- Key areas of declining or missed performance in 2021
- Uisce Éireann's overall progress against targets set for this revenue control period (2020-2024)

The section below provides summarises key areas reported for 2021, however further detail for each metric is provided in this report

- **Improved Performance:** The CRU welcomes the improvements delivered by Uisce Éireann in the following areas: bills issued based on a meter read to ensure non-domestic customers are accurately and fairly billed for their usage, customer satisfaction survey, and a reduction in the number of category 3-5 incidents that occurred.
- **Maintained Performance:** 100% of drinking sludge was disposed off in a satisfactory manner. Unresolved complaints upheld by the CRU CCT remained better than the average of those upheld against utilities or suppliers monitored.

Areas of Concern and Declining Performance: The CRU is concerned with the gap in the level of performance against the targets set.

- **Once off incidents relating to wastewater:** Uisce Éireann was to target 585 or less incidents for 2021 and less than 345 by 2024. 1,108 incidents were reported in 2020 and 1,067 incidents were reported in 2021.
- **Recurring incidents relating to wastewater:** Uisce Éireann was to target 170 or less incidents for 2021 and less than 98 by the end of 2024. 238 incidents were reported in 2020, 249 were reported in 2021.
- **Agglomerations with no wastewater treatment:** In 2021 Uisce Éireann was targeted to provide treatment at four sites. This would result in 31 sites remaining on the list at the end of 2021. Uisce Éireanne has reported 33 sites continue to remain discharging untreated wastewater into the environment at the end of 2021. The CRU

is concerned with the consistent delays in providing treatment at these agglomerations.

- **Customer Service and Reactivity:** A number of metrics related to Uisce Éireann's ability to respond to queries, complaints or issues within specified timeframes has dropped.

This includes failure to: provide a final decision on complaints within two months; restore supply after an unplanned interruption sooner than four or 12 hours; and to lift a boil water notice or drinking water restriction notice on water supplies within 30 days. The CRU is concerned with the drop in customer service performance reported in 2021, particularly given Uisce Éireann's ability to maintain its performance from 2019 into 2020. This is despite the initial challenges and uncertainties presented by the COVID-19 pandemic. The CRU is not satisfied with the reasons for reduced performance as set out in Uisce Éireann's submission.

- The CRU is concerned that Uisce Éireann only resolved 71% of its operations calls (versus target set at 80%)and 77% of its billing calls (vs target set at 90%) on the first call in 2021 which falls significantly short of the CRU's targets on both lines.
- Only 85% of the metered accounts that received a bill during the year received at least one bill based on a meter read which falls short of the CRU's target of 100%. This is the first year of reporting and the CRU expects that annual improvements can be delivered to meet performance targets.
- In 2021, Uisce Éireann delivered the lowest five-day response rate for both domestic and non-domestic complaints, since Uisce Éireann started reporting against this metric in 2018. This is also the case for final determinations issued within two months. The CRU is concerned with Uisce Éireann's continued declining trend in performance despite the substantial decrease in the number of complaints lodged with the utility. Further to compliance actions by the CRU, Uisce Éireann has committed to improving the customer complaints journey. The CRU therefore expects Uisce Éireann to take all steps to make the required improvements.
- In 2021, Uisce Éireann fell just short of its call abandonment target, achieving a rate of 4.4%. While the CRU considers Uisce Éireann to be in line with the 4% target in 2021, this represents a drop in performance compared to 2019 and 2020. This is of concern, if Uisce Éireann is to remain on track for the remainder of the RC3 period.

- Uisce Éireann answered and closed out 95% of billing contacts in 2021. While this performance is in line with the target, the CRU is very concerned that that Uisce Éireann's 2021 performance is the lowest in a declining trend in the four years reported to the CRU.

Quality of Water Supply

- Notwithstanding the welcome increase in lead compliance, the CRU shares the EPA's concerns with Uisce Éireann's progression in eliminating lead pipework overall. This is of particular concern in the context of the impending reduction of the lead compliance limit from 10µg/l to 5µg/l.
- While there have been improvements regarding microbiological parameters, including trihalomethane, the CRU is concerned with the decline in trihalomethane compliance overall.
- Uisce Éireann saw declining performance and did not meet the target for long-term notices Boil Water Notices in 2021 despite removing critical outstanding notices by the end of 2020.

Sewer incidents: Uisce Éireann has failed to begin reporting on internal and external sewer incidents despite confirming reporting capability for internal incidents as being in place since Q4 2019. Continued delays to reporting on this metric also impacts on the CRU's ability to assess performance and report on this and set meaningful and appropriate targets to improve Uisce Éireann's performance, ultimately prolonging customer issues. **Security of Supply:** The CRU is also concerned with continued delays to the National Water Resources Plan and to Uisce Éireann's reporting on security of supply.

Data Quality and Reporting:

Clear, accurate and transparent data from Uisce Éireann is key for the CRU to monitor Uisce Éireann's annual performance. Continued delays in Uisce Éireann providing data on a number of metrics, particularly those that have been in the Framework since 2016, remain a concern.

The reporting of new metrics, such as the reporting of sewer incidents and security of supply, are of key interest to customers in understanding which areas are most at risk and require targeted investment. Uisce Éireann's failure to provide this information to date hinders customers' ability to understand the utility's full performance. Providing this information could also improve customers' and stakeholders' input into consultations on proposed projects.

Next Steps

The CRU will continue to engage with Uisce Éireann to implement the remaining metrics included in the CRU's decision on the Framework during the RC3 period. This includes reviewing Uisce Éireann's customer side pipeage leakage data, implementation of the security of supply metric and revision of the customer satisfaction survey, among others. The CRU expects to see improvements from 2022 onwards as Uisce Éireann implements any necessary process improvements or reprioritisations to deliver against targets.

Public Impact Statement

This paper provides an overview of Uisce Éireann's performance to the end of 2021 on 27 metrics under six categories:

1. Customer service
2. Security of water supply
3. Quality of water supply
4. Sewer incidents
5. Environmental performance
6. Energy and emissions.

This paper is published to support openness, transparency and accountability on how Uisce Éireann is performing in terms of quality and value for money in the delivery of water and wastewater services to its customers.

Publishing this report helps to keep the public, and other key stakeholders, informed of Uisce Éireann's performance. Publicly reporting on delivery provides Uisce Éireann with a reputational incentive to deliver the outputs and outcomes expected of it. In addition, it promotes openness and transparency regarding the activities of Uisce Éireann and of the CRU. Monitoring also supports the CRU in making evidence-based decisions in the interest of customers.

The CRU considers Uisce Éireann's 2021 reporting submission is useful to understand how it is currently performing. With only two years of data submitted under the Framework in the RC3 period, trends are not fully evident. However, the CRU notes that many performance improvements are still required to get Uisce Éireann on track to meeting a number of targets by the end of the 2024 period.

There are a number of outstanding metrics for which Uisce Éireann has still not reported on under the Framework. Examples of these are the reporting of sewer incidents (i.e., areas where properties are impacted by wastewater flowing onto their property) and security of supply (i.e., which areas in the country are in a surplus of water or a deficit of water in a given year). These areas are of key interest to customers in understanding which areas are most at risk of experiencing unpleasant events such as wastewater entering a property or water restriction being put in place. Understanding the areas at risk also helps customers recognise where targeted investment is required. Uisce Éireann's failure to provide this information to date hinders customers' ability to understand the utility's full performance. Providing this information could also improve customers' and stakeholders' input into consultations on proposed projects.

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1. Introduction

1.1 The CRU Role

The CRU is Ireland's independent energy and water services regulator. Established in 1999, the CRU has a wide range of economic, customer protection and safety responsibilities.

The CRU is the regulator of Uisce Éireann as the national utility for the provision of public water and wastewater services. The CRU's role is to protect the interests of water and wastewater customers, ensure the delivery of water services in a safe, secure and sustainable manner and ensure that Uisce Éireann operates in an economic and efficient manner.

The CRU is responsible for setting the level of revenue that Uisce Éireann can receive, through various charges approved under the Water Charges Plan (new connections, non-domestic tariffs, etc) and via the Government subvention. The CRU does this by reviewing Uisce Éireann's submissions, engaging with the utility, benchmarking its proposed costs against comparator companies, completing a public consultation process and setting appropriate revenue allowances for operational costs, capital costs and other items. This process is known as a revenue control. The current revenue control, Revenue Control 3 (RC3) runs from the period 2020 to 2024.

Under RC3, CRU set out a number of outcomes that Uisce Éireann is required to deliver over the five years. Tied to these outcomes are an extensive and detailed set of outputs, covering projects and programmes across water and wastewater, that the CRU will monitor and hold Uisce Éireann accountable for delivering.

- Outcomes are the things that customer and society value, e.g. wholesome and clean drinking water;
- Outputs are specific things that the Uisce Éireann is required to deliver to (help to) achieve those outcomes, e.g. water treatment plants
- Inputs are the resources that Uisce Éireann will require to deliver those outputs

During each revenue control period, the CRU monitors Uisce Éireann's performance and delivery against the RC3 requirements. This includes monitoring Uisce Éireann's compliance with the CRU's Customer Handbooks, monitoring Uisce Éireann's delivery of its Investment Plan and assessing Uisce Éireann's performance against the metrics set out under the Performance Assessment Framework. The CRU reports on Uisce Éireann's delivery and performance based on the findings of its monitoring activities.

1.2 The Performance Assessment Framework

The CRU published its original decision on the Performance Assessment Framework that would apply to Uisce Éireann in November 2016.² The Framework provided a structured and clear basis for the CRU to assess Uisce Éireann's performance for its customers, supporting due oversight of the utility and evidence-based decisions by the CRU. The publication of reports under the Framework incentivises Uisce Éireann to improve its performance and service delivery for its customers and allows customers and other stakeholders of the utility to monitor that performance.

The Framework originally provided for the CRU's assessment of Uisce Éireann's performance across five categories: customer service, environmental performance, quality of water supply, security of water supply and wastewater (sewerage) service. Uisce Éireann submitted five reports to the CRU under the initial 2016 Framework. These are published alongside a CRU commentary or implementation report on the CRU's [website](#)³.

In 2021, the CRU reviewed the Framework to ensure the metrics remained appropriate in light of the CRU's Revenue Control 3 (RC3) decision. This review also included setting targets on Uisce Éireann's performance for the first time.

The revised Framework set out the categories and metrics, including standards of service, that the CRU requires Uisce Éireann to deliver for its customers. The CRU's monitoring of Uisce Éireann's progression and delivery of the Investment Plan, including associated outputs and outcomes, and of its compliance with the Domestic and Non-Domestic Customer Handbooks supports the CRU's assessment of the utility's overall performance under the Framework.

The CRU set targets for Uisce Éireann's performance over the period 2020 to 2024. While these targets were based on approved RC3 investment and the associated outputs and outcomes for that investment, among other factors, it is noted that the final performance target metrics were not confirmed until Q3 2021.

² [CER16308](#) Uisce Éireann Performance Assessment: Framework of Reporting Metrics

³ https://www.cru.ie/document_group/irish-water-performance-assessment/

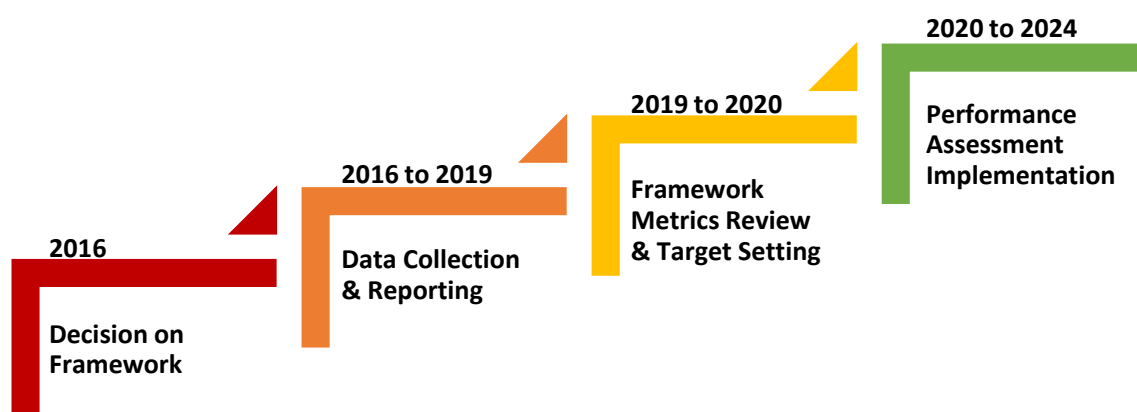


Figure 1 - Performance Assessment Implementation

The CRU set Uisce Éireann 2021 targets on 12 of the 27 metrics in the CRU's 2020-2024 decision on the Framework⁴. Of the 12 metrics with targets for 2021, a total of 19 targets were set against these (some metrics had more than one target).

This paper provides the second assessment of Uisce Éireann's performance against targets as set by the CRU under this Framework.

⁴ [CRU21101](#) Uisce Éireann Performance Assessment Framework Metric Review and Target-Setting 2020-2024 Decision Paper

1.3 Related Documents

- CRU202340a Uisce Éireann Performance Assessment Report No. 7
- CRU202340b Leakage Addendum to 2020 Monitoring Reports
- [CRU202228](#) Irish Water Performance Assessment Framework Annual Report 2020
- [CRU202228a](#) Irish Water Performance Assessment Report No. 6
- [CRU21101](#) Irish Water Performance Assessment Framework Metric Review and Target-Setting 2020-2024 Decision Paper
- [CRU20119](#) Irish Water Performance Assessment Framework Metric Review and Target-Setting 2020-2024 Consultation Paper
- [CER16308](#) Irish Water Performance Assessment: Framework of Reporting Metrics
- [CRU20140](#) Irish Water Performance Assessment 2019 Implementation Update
- [CRU20140a](#) Irish Water Performance Assessment Report No. 5
- [CRU19146](#) Irish Water Performance Assessment 2018 Implementation Update
- [CRU19146a](#) Irish Water Performance Assessment Report No. 4
- [EPA, 2022](#). Drinking Water Report in Public Supplies 2021
- [EPA, 2022](#). Urban Waste Water Treatment in 2021

Information on the CRU's role can be found on the CRU's website at www.cru.ie.

2. Uisce Éireann's Reported Performance

This section provides an overview of the information reported by Uisce Éireann for the year 2021, which is its second report to the CRU under the Performance Assessment Framework for 2020-2024.⁵ In cases where Uisce Éireann is not yet providing data regarding a metric, an update on the timelines for when this data is to be received is set out.

It is noted that the final performance target metrics were not confirmed until Q3 2021. As such, Uisce Éireann was not in a position to review or improve its processes to meet the 2020 or 2021 targets. However, these targets were calculated from allowed spend over RC3; previous Uisce Éireann performance; and the performance of comparator utilities in a similar timeframe. The CRU could reasonably expect Uisce Éireann to be at least close to the targets based on the above.

2.1 Customer Service

2.1.1 Ease of telephone contact – Speed of telephone response

Customers who contact Uisce Éireann via telephone are directed through an Interactive Voice Recognition (IVR) system. This IVR system directs customers through a series of menus and then places them in a queue to speak to an agent, thus enabling the company to route the call to the most appropriate agent with the intent of increasing efficiencies in call handling. This metric monitors the percentage of calls in the queue to speak to an agent that are answered within 20 seconds:

$$\frac{\text{Calls picked up by agent within 20 seconds in queue}}{\text{Total number of calls to Uisce Éireann}}$$

The CRU set Uisce Éireann a target of 85% under this metric for each of the years 2020-2024.

Uisce Éireann received 398,335 calls in 2021, compared to 357,042 in 2020 and 401,721 calls in 2019. Uisce Éireann answered 76% of calls within 20 seconds in the queue in 2021, which is a drop of 6% on its performance in both 2019 and 2020 (Figure 2).

The CRU is concerned with such a significant drop in performance, particularly given Uisce Éireann's ability to maintain its performance from 2019 into 2020, despite the initial challenges and uncertainties presented by the COVID-19 pandemic. It is disappointing to see the drop in performance was consistent across the twelve months of the year and the CRU is not satisfied with the reasons for reduced performance as set out in Uisce Éireann's submission.

⁵ [CRU/21/101](#) Uisce Éireann Performance Assessment Framework Metric Review and Target-Setting 2020-2024

Based on 2021 performance, the CRU no longer considers Uisce Éireann on track to meet the target of 85% for the coming years 2022-2024.

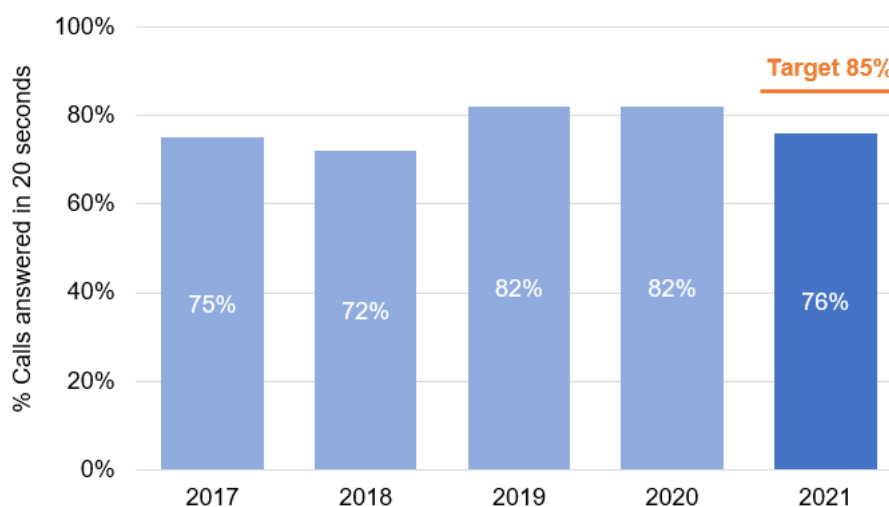


Figure 2 – Speed of Telephone Response

2.1.2 Ease of telephone contact – Call abandonment rate

Call abandonment rate monitors the percentage of calls abandoned in the queue to speak to an agent having been directed through the IVR system. Uisce Éireann is to target a call abandonment rate of 4% or less for each of the years 2020-2024.

In 2021, Uisce Éireann fell just short of its target, achieving a call abandonment rate of 4.4% (Figure 3). While the CRU considers Uisce Éireann to be in line with the 4% target in 2021, this represents a drop in performance compared to 2019 and 2020 and is a concerning trend if Uisce Éireann is to remain on track for the remainder of the RC3 period.

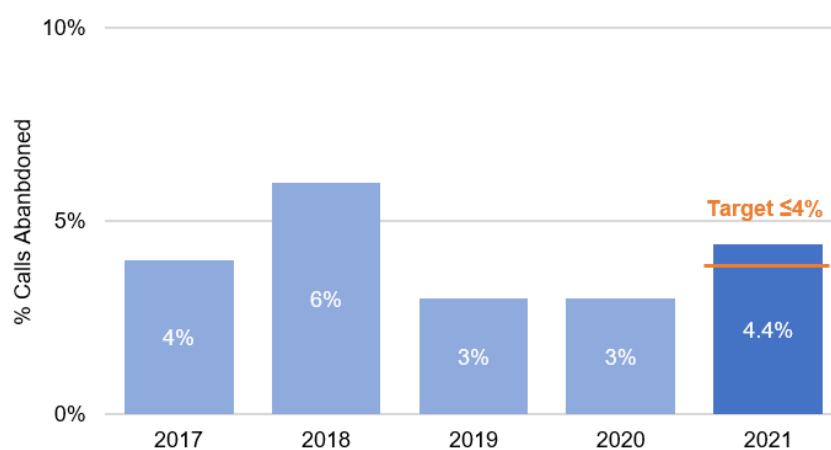


Figure 3 – Call Abandonment Rate

2.1.3 Ease of telephone contact – First call resolution

This metric is a measure of Uisce Éireann's ability to resolve the customer's query on the first call. Follow-up calls from the customer to Uisce Éireann within 30 days of the first call also result in the first call being treated as 'unresolved' under this metric. This can be calculated using the following:

$$\frac{\text{Number of calls to Uisce Éireann dealt with within one phone call}}{\text{Total number of calls to Uisce Éireann}}$$

Uisce Éireann is to target 90% of billing line calls and 80% of operations line calls to be resolved on the first call for each of the years 2020-2024.

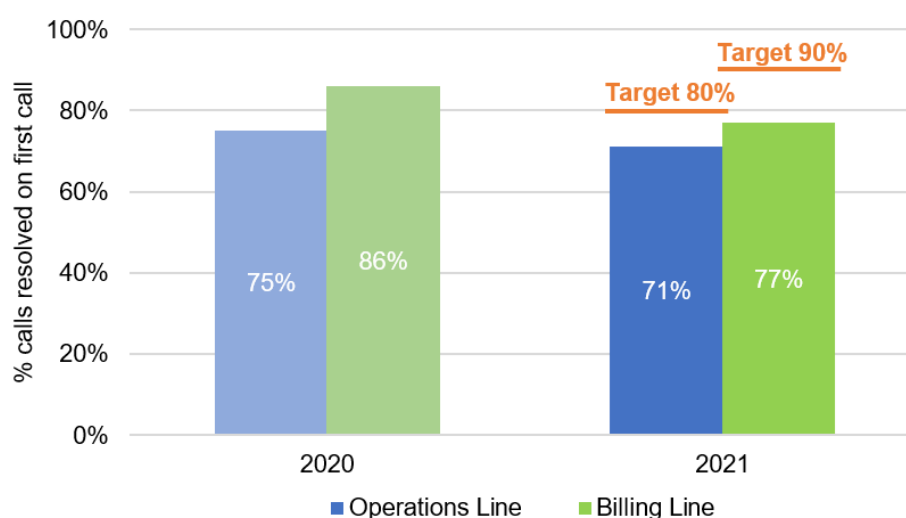


Figure 4 – First Call Resolution

Uisce Éireann resolved 71% of its operations calls and 77% of its billing calls on the first call in 2021. This performance falls significantly short of the CRU's targets on both lines in 2021.

In its 2020 Annual Report, the CRU considered that the required improvements to reach these targets by the end of the period were achievable by Uisce Éireann despite its failure to meet the targets in 2020. However, the CRU is now concerned with Uisce Éireann's ability to reach these targets as the metric has been in steady decline since 2018.

Uisce Éireann attributes the declining trend in its performance under this metric to a contact centre-wide trend in customers only calling companies to resolve more complex queries, with more straightforward queries often resolved through dedicated online channels.

However, the CRU would expect that given an increase in queries being resolved online, a similar trend of reduced calls to the contact centre over time should also be identifiable as only

the complex queries remain. As this trend is not yet evident, the CRU considers many less complex queries are being raised through calls that Uisce Éireann should be in a position to rectify on the call.

2.1.4 Billing of metered non-domestic customers

This metric only refers to non-domestic customers, as domestic customers do not receive bills from Uisce Éireann. The CRU monitors two sub-metrics relating to Uisce Éireann's billing of metered customers. The aim of this overall metric is to ensure that customers are accurately and fairly billed for their usage which, in turn, ensures Uisce Éireann appropriately recovers its service costs. In addition, improving the number of bills issued based on a meter read will also minimise the impact of 'bill-shock' on customers.

The CRU monitors the percentage of bills issued to metered accounts during the year that were based on a meter read. For the years 2020-2024, Uisce Éireann is to target 70% of bills issued to metered accounts during the year to be based on a meter read.

In 2021, 78% of the bills issued by Uisce Éireann to metered accounts were based on a meter reading. This is a significant increase compared to 68.6% in 2020 (see Figure 5). Uisce Éireann has surpassed its target of 70% of bills issued to metered accounts to be based on a meter read during the year.

The CRU welcomes the improvement here as it ensures that non-domestic customers are accurately and fairly billed for their usage and that Uisce Éireann can appropriately recover its service costs. This is particularly welcomed in 2021 as:

- a) due to COVID-19 restrictions, many non-domestic customers experienced changes in their water use
- b) The new harmonised non-domestic tariffs⁶ came into effect on 1st October 2021⁷

⁶ [CRU20072](#) Irish Water's Non-Domestic Tariff Framework – Tariff Application Rules Decision Paper

⁷ [CRU Press Statement](#): Irish Water Non-Domestic Tariffs Implementation Date Revised for a Second Time

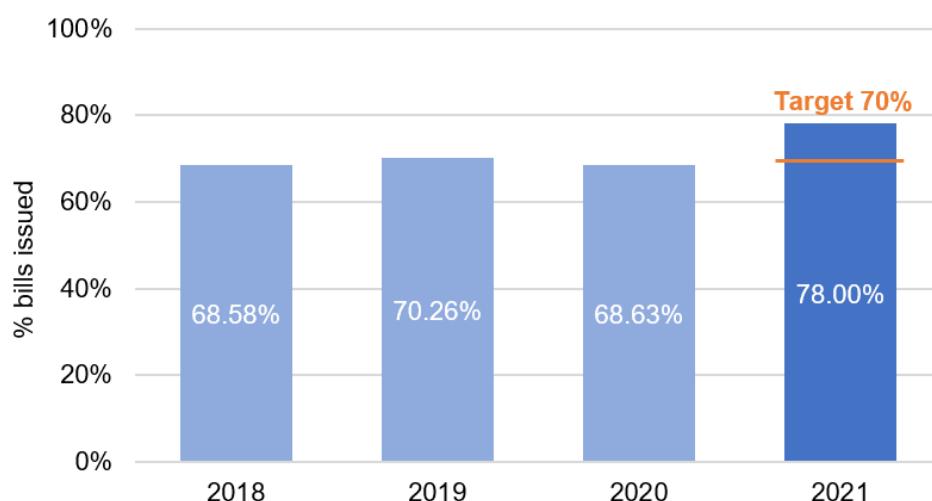


Figure 5 – Billing of Metered Customers

In addition to the sub-metric above, the CRU also monitors the percentage of metered accounts billed during the year that receive at least one bill based on a meter read. The CRU set Uisce Éireann a target of 100% under this new sub-metric.

Uisce Éireann provided data on this metric for the first time under this report. In 2021, 85% of the metered accounts that received a bill during the year received at least one bill based on a meter read. While this falls short of the CRU's target, the CRU notes that this is the first year of reporting and therefore expects with annual improvements, targets can be met by end of RC3.

Uisce Éireann noted in its submission that the metric itself does not account for instances where Uisce Éireann attempts to read a meter but cannot access it. However, in its request for information, the CRU set out that Uisce Éireann should outline where it has attempted to obtain a meter read which would have been used as the basis for a bill and could not do so. If Uisce Éireann provided such information, the CRU could consider this in relation to Uisce Éireann's performance. This information was not provided in Uisce Éireann's submission.

Based on Uisce Éireann's submitted data, the CRU considers Uisce Éireann is not currently on track to meet this target by the end of the RC3 period.

2.1.5 Response to billing contacts

Billing contacts refers to any communication from a customer that is related to bills. For example, this includes contacts regarding making a payment, changing billing details or meter readings.⁸

⁸ Billing contacts do not include complaints regarding bills. Billing complaints are captured as part of the *Response to Complaints* metric below.

This metric monitors the percentage of billing contacts that are answered and closed out within five working days. As with 2.1.4 *Billing of metered customers*, this metric includes billing contacts from non-domestic customers only, as domestic customers do not receive bills from Uisce Éireann. This metric will extend to include those from domestic customers who query bills in relation to Excess Use Charges⁹ once they take effect.

Uisce Éireann received approximately 134,400 billing contacts in 2021, which is a 32% decrease compared to approximately 198,000 billing contacts in 2020 and a 48% decrease on the 260,000 billing contacts in 2019. Uisce Éireann considers that reduced activities undertaken by Uisce Éireann during the COVID-19 pandemic may have continued to contribute to reduced calls in 2021.

Uisce Éireann answered and closed out of 95%¹⁰ of these billing contacts in 2021. While this performance is in line with the target the CRU is very concerned that that Uisce Éireann's 2021 performance is the lowest of the four years reported to the CRU (see Figure 6).

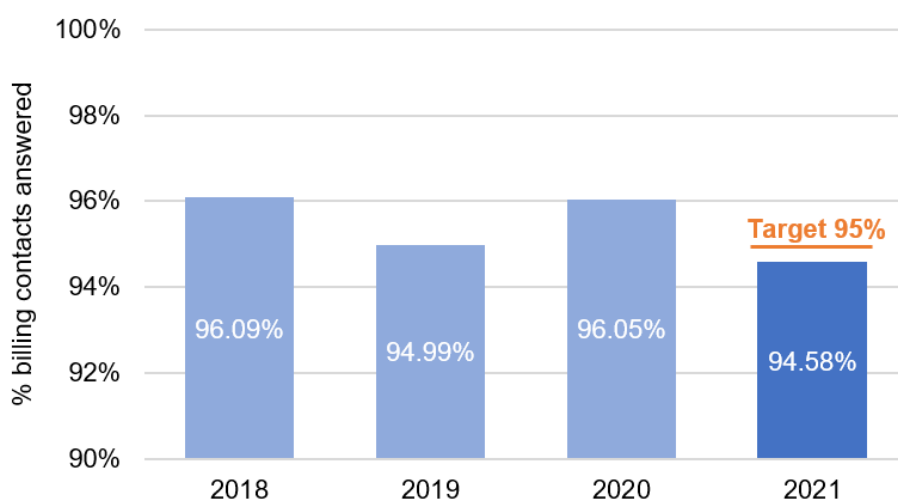


Figure 6 – Response to Billing Contacts

⁹ [CRU/19/086](#) Uisce Éireann's Household Water Conservation Proposal (Excess Use Charges) CRU Decision and Response to Comments Paper

¹⁰ Rounded up from 94.58%

2.1.6 Response to complaints

The CRU's Domestic and Non-Domestic Customer Handbooks¹¹ set out the definition of a complaint as follows: the expression (through various possible channels, for example: letter, email, phone call, physical claim) of a customer's dissatisfaction and their explicit expectation for a response or resolution. This metric monitors Uisce Éireann's ability to respond to and close out complaints within the defined timeframes set out in these Handbooks.

Uisce Éireann is to target the following under this metric for each year 2020-2024:

- 100% of complaints to be responded to within five working days, either with a resolution or an outline plan of the proposed resolution
- 100% of complaints to be issued a final decision within two months

Uisce Éireann received just under 6,500 complaints in 2021, a 28% decrease on 9,000 complaints in 2020 and a 54% decrease on over 14,000 complaints received in 2019.

Uisce Éireann's response to complaints is broken down into domestic and non-domestic complaints.

In 2021, Uisce Éireann responded to 94.45% of domestic and 90.04% of non-domestic complaints within five working days, with either a resolution or an outline plan of the proposed resolution (Figure 7). This is the lowest five-day response rate for both domestic and non-domestic complaints since Uisce Éireann started reporting against this metric in 2018.

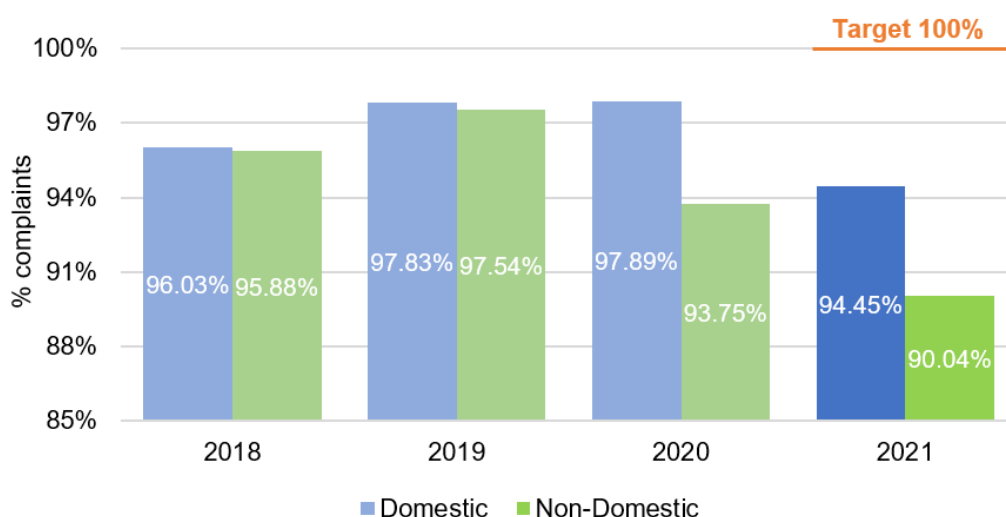


Figure 7 – Response to Complaints Five Working Days¹²

¹¹ The CRU's [Domestic](#) and [Non-Domestic](#) Customer Handbooks and related decisions are available on the CRU website.

¹² Data is provided from Q2-Q4 in 2018 and from Q1-Q4 in each of the years 2019-2021.

Uisce Éireann issued a final decision within two months to 95.74% of domestic and 88.16% of non-domestic customers in 2021 (Figure 8).

As with the five-day sub-metric above, Uisce Éireann's 2021 performance regarding complaints raised by non-domestic customers is, for the second year in a row, the lowest since it first reported in 2018.

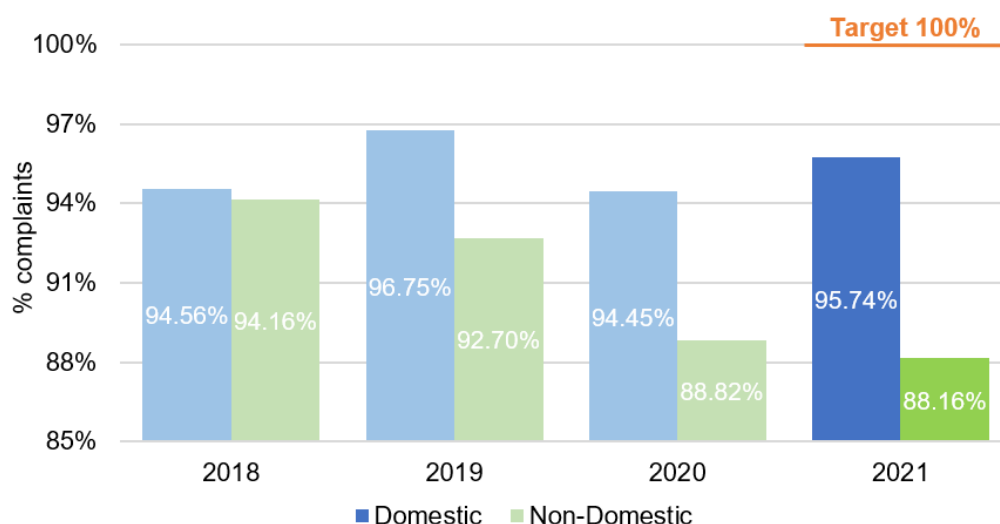


Figure 8 – Response to Complaints Two Months

The CRU is concerned with Uisce Éireann's continued declining trend in performance despite the substantial decrease in the number of complaints lodged with the utility. The CRU is not satisfied with the reasons for reduced performance as set out in Uisce Éireann's submission. Resolving complaints in a timely manner is a critical customer service provision for customers to ensure that where issues have occurred they are rectified swiftly. The CRU is concerned with Uisce Éireann's performance in this matter and expects to see improved service and performance metrics in future reports.

Uisce Éireann's *Response to Complaints* performance is critical for customers who remain dissatisfied after their complaint has been closed by Uisce Éireann, as the customers will have been through Uisce Éireann's full complaints process and issued with a final decision before they can raise a complaint to the CRU. Where Uisce Éireann is not meeting the target to issue a final decision within two months, it prolongs the difficulty being faced by the customer.

In addition to the concerns with Uisce Éireann's performance in responding to complaints, the CRU also has ongoing concerns regarding Uisce Éireann's classification of complaints.

In 2021, the CRU investigated an issue in Bailieborough, Co. Cavan, which took place in late 2019.¹³ One of the CRU's findings was that of 96 'contacts' received by Uisce Éireann in relation to the Bailieborough water supply, the majority should have been categorised as complaints.

In 2022, the CRU investigated issues in Cavanhill, Co. Louth, and Gorey, Co. Wexford which took place in 2021.¹⁴ Again, the CRU found that Irish Water did not accurately categorise several contacts as complaints despite customers expressing dissatisfaction and an expectation of a response or resolution.

These instances had implications for how contacts were handled and resolved, and for comparative customer service metrics, such as the response to complaints and unresolved complaints metrics that are monitored as part of this Framework.

The CRU has requested a number of remedial actions to address the process failings which led to the issues in Bailieborough, Cavanhill, and Gorey, including a review of processes to categorise customer contacts to ensure that complaints are captured appropriately. One of the actions, included the requirement to review against comparable water utilities to identify best practice and potential improvements in Uisce Éireann's processes. The CRU is liaising with Uisce Éireann to ensure remedial actions are addressed satisfactorily and will continue to monitor Uisce Éireann's compliance with its legislative obligations as set out in the approved Codes of Practice¹⁵.

Through engagement with Uisce Éireann to improve its complaints process, the CRU notes that, during 2022, Uisce Éireann reviewed some of its complaints-handling practices such as revising complaints training material in conjunction with the Contact Centre and holding refresher trainer sessions.

Uisce Éireann stated that its senior management attended the Contact Centre and presented on the overall complaints process to ensure advisors were familiar with the full end-to-end journey and escalations process. Additionally, it put in place increased headcount to manage complaint handling processes and implemented and continues to work on any remedial actions arising from audits and investigations carried out during this period. The CRU welcomes Uisce Éireann's commitment to improving the customer complaints journey and will report on Uisce Éireann's performance for 2022 in the next Annual Report.

¹³ [CRU21128](#) Compliance Investigation Uisce Éireann – Bailieborough Public Water Supply

¹⁴ [CRU202294](#) Compliance Investigations: Irish Water – Cavanhill Public Water Supply & Gorey Regional Creagh Public Water Supply Information Note

¹⁵ The approved Codes of Practice are available on Uisce Éireann's website [here](#).

2.1.7 Unresolved Complaints Upheld by CRU Customer Care Team

This metric monitors the total number of unresolved complaints upheld by the CRU Customer Care Team (CCT). The aim of this metric is to provide further information regarding Uisce Éireann's overall complaints handling, outside of Uisce Éireann's ability to respond to complaints (see section 2.1.6 above). Further information on Uisce Éireann's performance is publicly available in the CRU's Customer Care Team Annual Reports.¹⁶

Uisce Éireann's target is for the number of its complaints upheld by the CRU's CCT to be better than the average of either

- a. those upheld in relation to other regulated network utilities when calculated to account for the number of sites served, or
- b. those upheld in relation to energy suppliers when calculated to account for the customer base.

For the purposes of comparison, the number of sites or customer base for network companies, suppliers and Uisce Éireann is calculated using the number of individual connections¹⁷.

$$\frac{\text{Number of complaints upheld}}{\text{Number of connections}} \times 10,000$$

The CRU CCT upheld 8 complaints against Uisce Éireann in 2021.¹⁸ As Uisce Éireann had 2,424,172 individual connections in 2021, this equals 0.03 complaints upheld per 10,000 connections, an increase on 0.02 complaints upheld per 10,000 connections in 2020.

The average number of complaints upheld against the 9 suppliers used in the CCT reports was 0.34 complaints per 10,000 customers. In terms of the network companies, an average of 0.1 complaints were upheld against ESB Networks and Gas Networks Ireland per 10,000 sites.

As Uisce Éireann's average complaints per 10,000 connections (0.03) is lower than that of the averages of suppliers and network companies (0.34 and 0.1), Uisce Éireann achieved the target level of performance for this metric in 2020.

¹⁶ The CRU CCT publishes annual reports regarding its complaint resolution service which are available on the CRU website [here](#).

¹⁷ The number of individual connections differs from the number of connected properties as some properties may have more than one individual connection.

¹⁸ [CRU202243](#) Customer Care Team Annual Report 2021

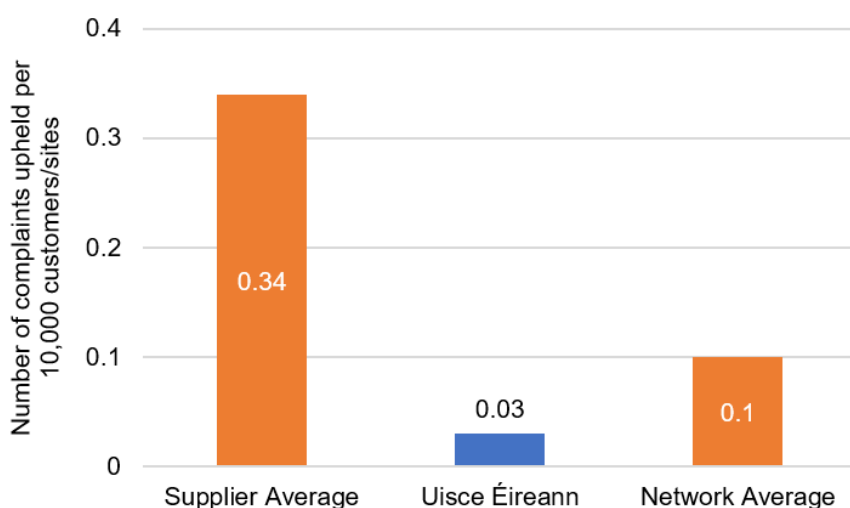


Figure 9 – Number of complaints upheld per 10,000 customers/sites in 2021

In addition to complaints handling, Uisce Éireann is required to issue charter payments where it fails to meet its commitments as outlined in its Domestic Customer Charter¹⁹.

In 2021, the charter payment increased from €10 to €30. Uisce Éireann made 84 charter payments in 2021, more than double the 32 charter payments made in 2020. This includes payments where Uisce Éireann recognised the breach itself and payments which were directed by the CRU to be made by Uisce Éireann.

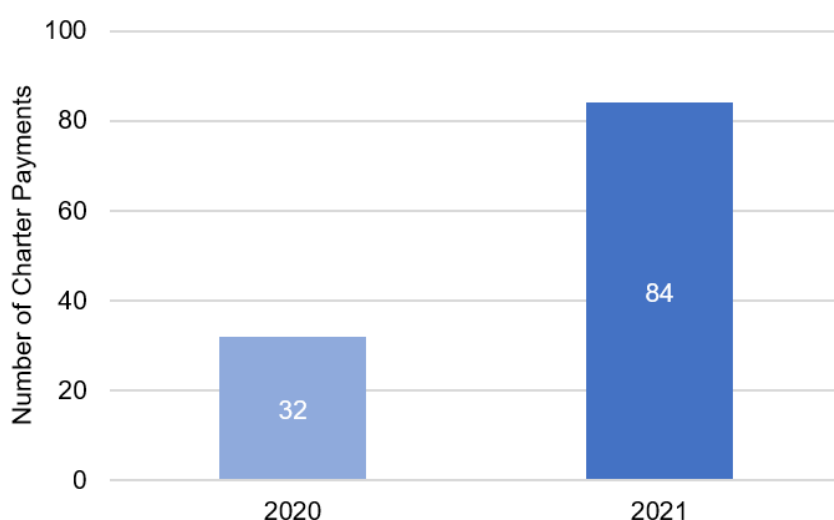


Figure 10 – Number of charter payments made by Uisce Éireann

¹⁹ Uisce Éireann [Domestic Customer Charter](#)

The CRU's CCT Annual Report also outlines the types of contacts the CRU receives from customers during the year.

In 2021, Uisce Éireann again drove the fourth largest percentage of contacts from a supplier or network operator, accounting for about 6% of the total contacts received by the CRU that year.

Customers continued to note that Uisce Éireann was failing to respond to their queries in a timely manner and/or not escalating or responding to customer complaints. Again, this impacts on the CRU's monitoring of complaints related metrics under the Framework.

2.1.8 Customer Satisfaction Survey

Customer satisfaction with Uisce Éireann is determined by a survey of customers who have contacted Uisce Éireann via telephone during the year. This satisfaction relates only to the telephone contact itself, not overall satisfaction with Uisce Éireann's service provision. It is presented as a customer satisfaction (CSAT) score, which is calculated from the surveyed customers who rated as satisfied (7-10) on a ten-point scale.

As set out in the CRU's decision paper on the Framework for 2020-2024, Uisce Éireann is to target an increasing trend under the original survey as set up in 2016 for the years 2020-2022. Uisce Éireann is to engage with the CRU and an independent research company during 2022 as necessary to facilitate the commencement of an updated survey²⁰ in January 2023. For the years 2023-2024, Uisce Éireann is to establish a performance baseline under the new survey.

Uisce Éireann achieved a CSAT score of 78% in 2021, increasing on the 77% achieved in 2020. This means Uisce Éireann met its target of an increasing trend in performance (Figure 11).

²⁰ The updated survey is intended to better reflect the purposes of this metric and that the methodology, including the approach to scoring, is in line with best practice. The CRU also aims to expand this survey to monitor the satisfaction of all customers who contact Uisce Éireann, for example via social media or email, when the rate of customer engagement with these surveys is at an appropriate level.

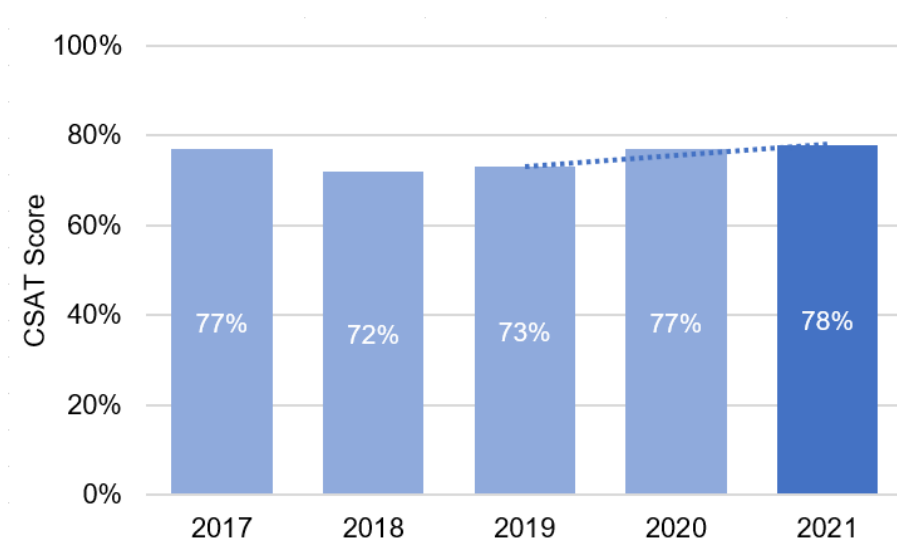


Figure 11 – Customer Satisfaction (CSAT) Score

2.1.9 Stakeholder Engagement

The CRU decided to set up a stakeholder panel on a pilot basis during the RC3 period to assess Uisce Éireann's stakeholder engagement. The CRU will not set targets for performance under this metric during this pilot phase. The CRU will review the operation of the panel and consider targets for the RC4 period towards the end of the RC3 period.

It is envisaged that the stakeholder panel will be comprised of members that represent key Uisce Éireann stakeholders and stakeholder groups. The stakeholder representatives will score Uisce Éireann's stakeholder engagement based on their experience in relation to a defined number of engagements each year as outlined by the CRU. This may include, for example, Uisce Éireann's engagement with stakeholders when carrying out planned consultations and policy implementation and during incident management and recovery.

Work to establish the panel, including drafting terms of reference, determination of membership and setting the basis for assessing Uisce Éireann's performance will be determined after engagement with stakeholders and Uisce Éireann, is currently scheduled to commence in 2023. As such, the CRU expects the pilot phase of the stakeholder panel to continue in the early years of RC4 (2025-2029). The experience during the pilot phase of the stakeholder panel will be considered by the CRU in advance of the decision to formally assess Uisce Éireann's performance using the panel. The CRU will provide an update on the establishment of the panel in the next report on Uisce Éireann's performance.

2.2 Security of Water Supply

2.2.1 Security of Supply

The CRU intends to monitor two sub-metrics to report on security of supply: an overall Security of Supply Index, and the number of water resource zones in deficit and the population served by those resource zones. Uisce Éireann has not reported on these metrics to date, citing the need for full implementation of a National Water Resources Plan in order to begin reporting.

In June 2021, Uisce Éireann published the Framework Plan for its National Water Resources Plan (NWRP). This plan sets out how Uisce Éireann intends to provide a safe, secure, and reliable water supply over the next 25 years, while considering any impacts on the environment and the impact of climate change, growth, and future demand. The full NWRP identifies any existing and potential shortfalls in meeting water demand under normal and critical weather conditions, based on a defined methodology and level of service, in all 539 of Uisce Éireann's water resource zones.

The CRU set interim targets for Uisce Éireann to publish its first NWRP Framework Plan and the subsequent four Regional Water Resources Plans by the end of 2021, to support Uisce Éireann's future reporting under this metric.

In June 2021, Uisce Éireann published the NWRP Framework Plan, meeting part of the target set by the CRU. However, Uisce Éireann published only the first of the four Regional Water Resource Plans for consultation in December 2021, and so ultimately did not meet the CRU's interim target to publish all four decisions by the end of 2021.

The CRU intended to engage with Uisce Éireann in 2022 to better understand the methodology set out in the National Water Resources Plan Framework Plan to facilitate reporting under this metric for the first time in 2022 (providing 2021 data). As Uisce Éireann maintains that all aspects of the National Water Resources Plan must be decided and in place before reporting on the Security of Supply metrics can begin, this engagement did not occur. The CRU is concerned with continued delays to the National Water Resources Plan and to Uisce Éireann's reporting on security of supply.

As mentioned in the 2020 Annual Report, the CRU was open to considering the inclusion of a third sub-metric, a 10-year capacity register, as proposed by Uisce Éireann during the consultation process of the Framework for 2020-2024. In order to consider the proposed sub-metric, the CRU required Uisce Éireann to make a submission regarding its proposed 10-year capacity register in Q1 2022 for CRU consideration, including consideration of how it relates to the other two methods outlined above. The CRU received this submission by the stated deadline

and ultimately determined that, while useful for Uisce Éireann's internal use, this method was not suitable for inclusion in the Framework. The CRU considered the proposed method did not align with the other methods currently included in the Framework. The CRU's final decision on this matter was made and communicated to Uisce Éireann in June 2022.

2.2.2 Leakage

This metric is designed to monitor the amount of water lost on the public network and the amount of water lost on customer supply pipes, separately. The CRU set Uisce Éireann a target to reduce leakage by 161 million litres per day on the public side and by 15 million litres per day on the customer supply pipe.

As per the Irish Water Performance Assessment 2020 to 2024 Metric Review and Target Setting²¹ and Irish Water Revenue Control 3 – Financial Incentives (Non-Domestic Billing and Leakage)²² decision papers, Uisce Éireann is required to submit detailed information on leakage at both the public and private (customer supply pipe) side for the purposes of Investment Plan monitoring²³, the Performance Assessment Framework monitoring and the calculation of the RC3 Financial Incentives allowance.

Despite extensive engagement on this issue Uisce Éireann had not provided the required information on public side and customer supply pipe leakage in the correct format or in sufficient detail to enable the CRU to report against it in the 2020 Annual Report. The CRU wrote to Uisce Éireann to request this data be provided under Section 43 of the Water Services (No. 2) Act 2013.²⁴ On receipt and review of the data, the CRU is publishing, alongside this report, a leakage addendum²⁵ to the Performance Assessment Framework 2020 Annual Report and the Capital Investment Plan Monitoring Report for 2020. This addendum outlines Uisce Éireann's public side leakage performance in 2019 and 2020, as well as outlining the reasons for the delay in receiving sufficient data from Uisce Éireann.

In order to assess and report on public side leakage in a clear and transparent manner, the CRU requests data be provided in the form of a Water Balance (Figure 12, see Appendix B for more detail). To complement the Water Balance, the CRU requests methodology statements also be submitted, which outline the assumptions and estimations made by Uisce Éireann when calculating the Water Balance. Methodology statements are key for understanding where Uisce

²¹ [CRU21101](#) Irish Water Performance Assessment Framework 2020 to 2024 Metric Review and Target Setting

²² [CRU21108](#) Irish Water Revenue Control 3 – Financial Incentives (Non-Domestic Billing and Leakage)

²³ Further information on the CRU's monitoring of Uisce Éireann's capital Investment Plan is available on the CRU's website [here](#).

²⁴ [CRU202229](#) CRU Letter to Uisce Éireann Leakage Request for Information

²⁵ CRU202340b CRU Leakage Addendum to 2020 Monitoring Reports

Éireann has improved its data or assumptions, which can have an impact on the reported performance.

Highlighting adjustments to Uisce Éireann's data improvements ensures that changes in its leakage performance are correctly attributed to Uisce Éireann's efforts to reduce leakage rather than to the improved data. This also means that any updates made to the methodology statement may result in the updated year no longer being comparable to the preceding years that have not made the change.

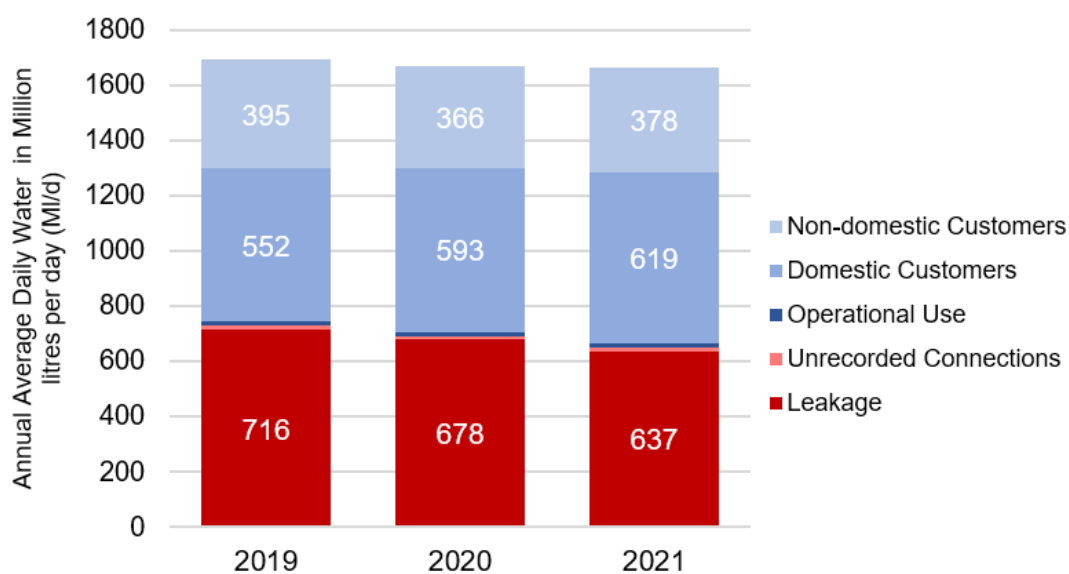


Figure 12 – Water Balance

In 2021, Uisce Éireann estimated that on average 637 million litres of water per day (MI/d) were lost on the public network. As no changes were made to its methodology, this figure is directly comparable to the 2019 and 2020 reported years. Therefore, to date, Uisce Éireann has reduced leakage by 79 MI/d since 2019 (see Figure 13).

The CRU considers Uisce Éireann is currently on track to meeting the public side leakage reduction target of 161 MI/d by the end of RC3.

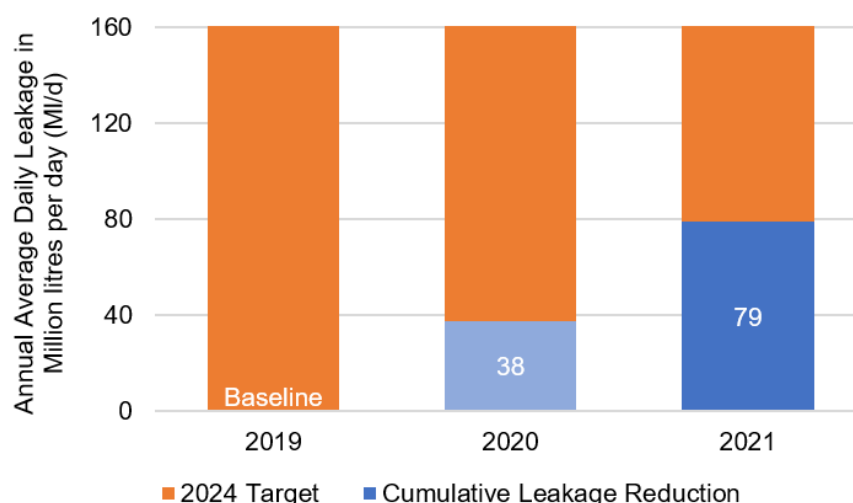


Figure 13 – Cumulative Annual Leakage Reduction

Uisce Éireann provided its first submissions regarding customer supply pipe leakage in December 2022. The CRU is currently reviewing this data and will provide an update in future reports. The CRU welcomes this submission, however, remains concerned with the one-year delay in providing a baseline estimate for customer supply pipe data (i.e., 2019 data) despite its inclusion as a targeted outcome in the CRU's RC3 decision paper²⁶ and RC3 Financial Incentives (Non-domestic Billing and Leakage) decision paper²⁷.

²⁶ [CRU19148](#) Irish Water Revenue Control 3 Decision Paper

²⁷ [CRU21108](#) Irish Water Revenue Control 3 – Financial Incentives (Non-Domestic Billing and Leakage)

2.3 Quality of Water Supply

2.3.1 Interruptions to supply

The 2016 Framework included a metric to monitor the number of properties subject to unplanned interruptions divided into three time-bands:

- interruptions lasting for greater than 4 hours,
- interruptions lasting greater than 12 hours, and
- interruptions lasting greater than 24 hours.

Uisce Éireann is not currently in a position to capture the exact number of properties experiencing unplanned interruptions to their supply during each outage. To address this, Uisce Éireann applies an approach whereby, it estimates the greatest number of properties impacted by a particular interruption, to ensure vulnerable customers are captured and notified or provided with an alternative supply where needed. In order to estimate this figure, Uisce Éireann takes a description of the area provided by Local Authorities and interprets the information via Geographic Information System mapping (a mapping tool which enables it to see its network).

The CRU will replace monitoring unplanned interruptions that last for greater than 4 hours with monitoring the minutes of lost supply arising from both planned and unplanned interruptions. This will allow for benchmarking of Uisce Éireann against water companies in neighbouring jurisdictions. The CRU required Uisce Éireann to set out when it will first report under this metric and an associated plan by the end of Q1 2022. This plan was submitted on time and outlined that Uisce Éireann expects to first report on this metric in 2024 (2023 data). In the interim to this change, the CRU will continue to provide information on the 'greater than 4 hours' time-band.

In 2021, out of all properties connected to Uisce Éireann's network, Uisce Éireann estimated that:

- 65.9% of properties experienced an unplanned interruption for greater than 4 hours,
- 25.3% experienced an unplanned interruption for greater than 12 hours, and
- 6.4% experienced an unplanned interruption for greater than 24 hours.

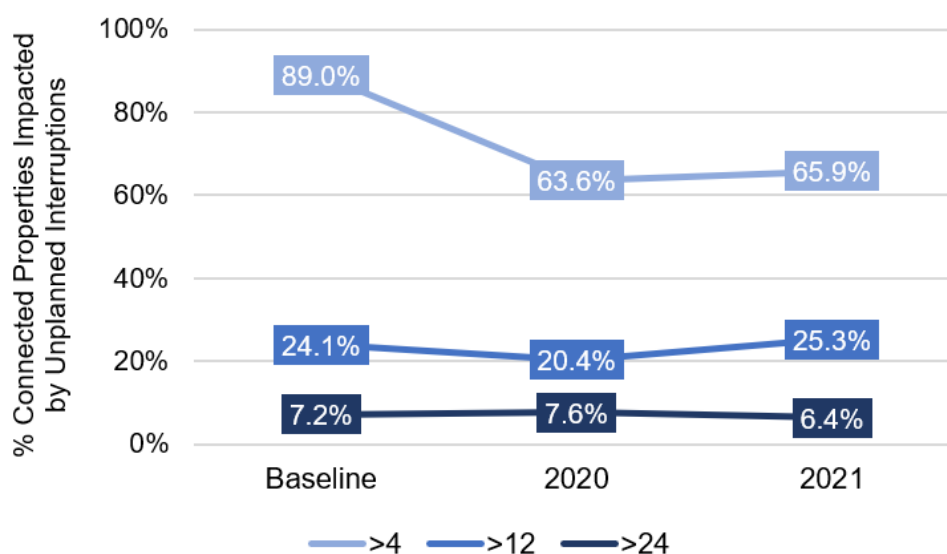


Figure 14 – Percentage of Connected Properties Impacted by an Unplanned Interruption to Supply²⁸

While performance under greater than 4 and 12 hours declined since 2020, there were less properties estimated to have been impacted by an unplanned interruption for greater than 24 hours in 2021.

The CRU did not set annual targets for Uisce Éireann under this metric. Instead, the CRU placed targets on Uisce Éireann's performance to the end of the RC3 period, end 2024:

- For greater than 4 hours, the CRU set a 'decreasing trend' target under the sub-metric until such time as Uisce Éireann begins reporting on the replacement metric, minutes for lost supply. As the 2021 performance has improved on the baseline, Uisce Éireann continues to meet this target.
- For greater than 12 hours, the CRU set Uisce Éireann a target of less than 12% of connected properties to be impacted by end 2024. Uisce Éireann is no longer on track to meet that target as its performance has now fallen below that of the baseline performance.
- For greater than 24 hours, the CRU set Uisce Éireann a target of less than 3.6% of connected properties to be impacted by end 2024. Based on the improvement made in 2021, the CRU considers Uisce Éireann is currently on track to meet this target.

While the CRU agrees with Uisce Éireann's aim to ensure all potentially impacted vulnerable customers are catered for appropriately, the CRU also considers that Uisce Éireann is in a

²⁸ As reporting capability for unplanned interruptions to supply has only been in place since Q2 2019, the CRU used Q3 2019 to Q2 2020 data as an annual average for establishing a baseline.

position to enhance this reporting and refine the accuracy of its data for reporting the number of properties impacted by an unplanned interruption. This is considering that Uisce Éireann makes its data submission in Q3 of the subsequent year and so can improve its understanding of the properties likely to have been impacted over time.

Uisce Éireann is to provide both the 'greatest estimated number of properties' figure as currently provided in its submissions and its 'revised estimated number of properties' annually for each metric reported to the CRU.

A target is also set for Uisce Éireann to provide the CRU with its plan to refine the estimated number of properties affected by an unplanned interruption by the end of 2021 to facilitate reporting the 'revised estimated number of properties'. Uisce Éireann submitted its plan and therefore has met this target. In its submission Uisce Éireann, stated that ²⁹ and related data over Q1 to Q3 2022. Uisce Éireann intends to engage further with the CRU regarding the outcomes of its review, including improvements identified. An overview of these outcomes and improvements will be outlined and an update provided in the 2022 Annual Report.

2.3.2 Drinking Water Quality

Uisce Éireann is responsible for ensuring drinking water meets the quality standards set out in the Drinking Water Regulations, enforced by the EPA. Uisce Éireann submits its water quality results to the EPA, and the EPA audits Uisce Éireann's monitoring to ensure it is fit for purpose. Where there is a water quality failure, the EPA oversees Uisce Éireann's investigation and action. Where there is a failure to meet these standards, or where there is a public health risk, Uisce Éireann must consult with the Health Service Executive (HSE).

The Framework includes metrics regarding Uisce Éireann's performance across six drinking water quality metrics:

- overall microbiological compliance and *E. coli* and Enterococci compliance, and
- overall chemical compliance and lead and trihalomethane compliance.

The EPA's Drinking Water Quality in Public Supplies 2021 report³⁰ notes that sample compliance rates remain consistently high year to year. In 2021, overall microbiological compliance was 99.96% and overall chemical compliance was 99.60%.³¹

²⁹ Geographic Information System

³⁰ [EPA, 2022](#). Drinking Water Report in Public Supplies 2021

³¹ The CRU has used the EPA's verified figures here instead of those included in Uisce Éireann's submission.

The high overall microbiological compliance is a result of continued high compliance with the two parameters within this category. In 2021, compliance with the *E. coli* parameter was 99.95% and compliance with the Enterococci parameter was 100%.

Trihalomethanes are formed when organic matter in water reacts with chlorine used in the disinfection process. Figure 15 below shows that there was a decline in trihalomethane compliance, with 95.30% in 2021 compared to 97.34% in 2020.

Lead is found in drinking water if it dissolves from lead pipework, fittings and connections. In 2017, Uisce Éireann estimated that there were approximately 180,000 residential properties at risk of lead non-compliance.³² Of this, 40,000 were backyard lead service connections which loop off the mains and run through backyards serving several houses. The remaining 140,000 were individual lead service connections from water mains which run under the roads. By the end of 2021, Uisce Éireann had removed a total of approximately 41,000 lead service connections under the Lead in Drinking Water Mitigation Plan and treated a further 1,607 by dosing orthophosphate.

Compliance with the lead parameter, which is 10 micrograms per litre (µg/l), fluctuated between 97.8% and 98.9% between 2014 and 2020. In 2021, compliance with the lead parameter was 98.92%. While Uisce Éireann is making progress to improve its compliance with the lead parameter, the EPA maintains concerns around its ability to fully comply with the lead parameter and remove all lead pipework as soon as possible. In its Drinking Water Quality in Public Supplies 2021 report, the EPA stated:

“Progress by Irish Water on individual lead connections recovered somewhat during 2021. However, this progress rate is still unacceptable and it will take 24 years to address the risks posed to public health from lead in drinking water at the replacement rate observed in 2021 which is a significant overrun of Irish Waters’ original plan for completion in 2026. Additionally, the Department of Housing, Local Government and Heritage and the Department of Health have not yet published a progress report on the National Lead Strategy. The compliance limit for lead is set to be halved in 2036 which increases the challenge that needs to be addressed. Given the cumulative risk to human health posed by any level of lead in water supplies and the forthcoming reduced lead limit in the new EU Drinking Water Directive, leadership at a national level is required to address this urgent matter.”

³² Uisce Éireann [Lead in Drinking Water Mitigation Plan](#)

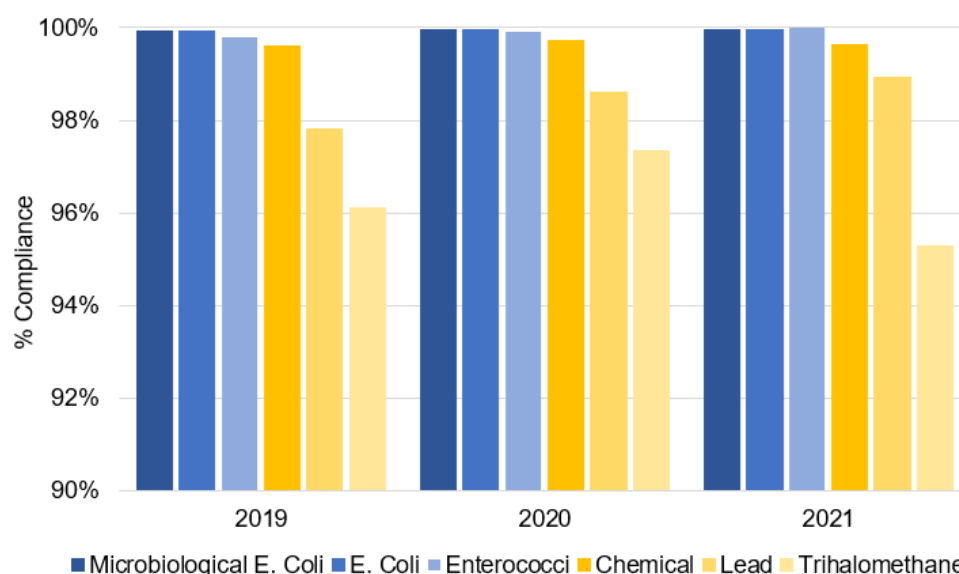


Figure 15 – Drinking Water Quality

The CRU shares the EPA’s concerns with Uisce Éireann’s progression in eliminating lead pipework overall. This is of particular concern in the context of the impending reduction of the lead compliance limit from 10µg/l to 5µg/l.

However, Uisce Éireann’s consistent high compliance with the microbiological parameters, as well as overall chemical compliance are positive achievements. The CRU also welcomes Uisce Éireann’s increased lead compliance after a drop in 2019.

While there have been improvements regarding the number of people affected by persistent trihalomethane failures due to investment made in the Vartry water treatment plant, the CRU is concerned with the decline in trihalomethane compliance overall. This decline is especially notable given the European Commission is progressing infringement proceedings against Ireland for failure to address trihalomethane compliance.

From Uisce Éireann’s performance in 2021, the CRU considers Uisce Éireann is currently on track to meet its 2024 minimum targets as set by the CRU for each of the parameters except trihalomethane compliance due to its decline on 2020 performance.

2.3.3 Water Supplies on Boil Water Notices and Water Restrictions

Boil water notices and/or drinking water restriction notices are issued by Uisce Éireann in consultation with the HSE when drinking or using water supplied by the utility might endanger people's health.

Monitoring these notices is a reflection of Uisce Éireann's service provision and reflects Uisce Éireann's work to remove long-term³³ notices and Uisce Éireann's investment in reducing the risk of new notices being issued.

In 2021, 46,069 people on 29 supplies were impacted by a boil water notice for greater than 30 days. This is three times the amount of people impacted in 2020, where 15,493 people were impacted by boil water notices on 26 supplies.

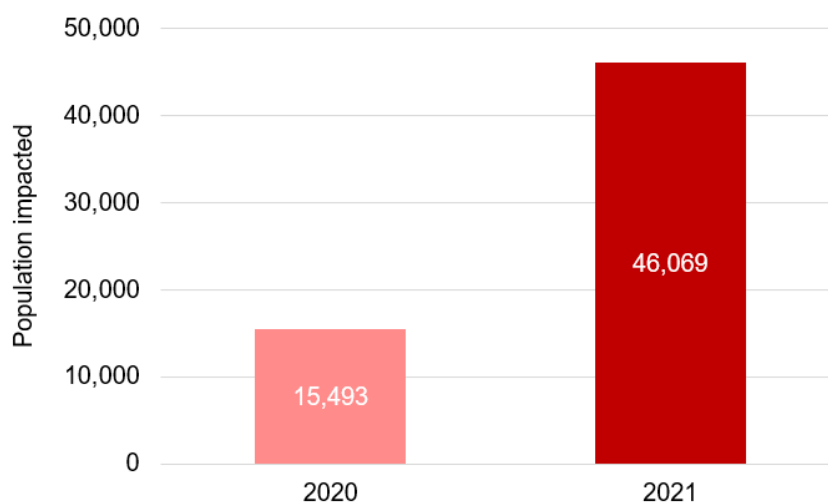


Figure 16 – Population impacted by long-term boil water notices³⁴

In addition to the boil water notices, 1,208 people were affected by 9 supplies issued a drinking water restriction notice for greater than 30 days. Again, this is an increase on 952 people affected by 8 supplies in 2020.

³³ The EPA distinguishes between long-term (in place for greater than 30 days) and short-term (in place for less than 30 days) boil water and drinking water restriction notices.

³⁴ Uisce Éireann has advised the CRU previously that the numbers presented above differ from those published by the EPA as clarification is required over responsibility for several small supplies. The EPA has included these supplies in its reports, Uisce Éireann has not.

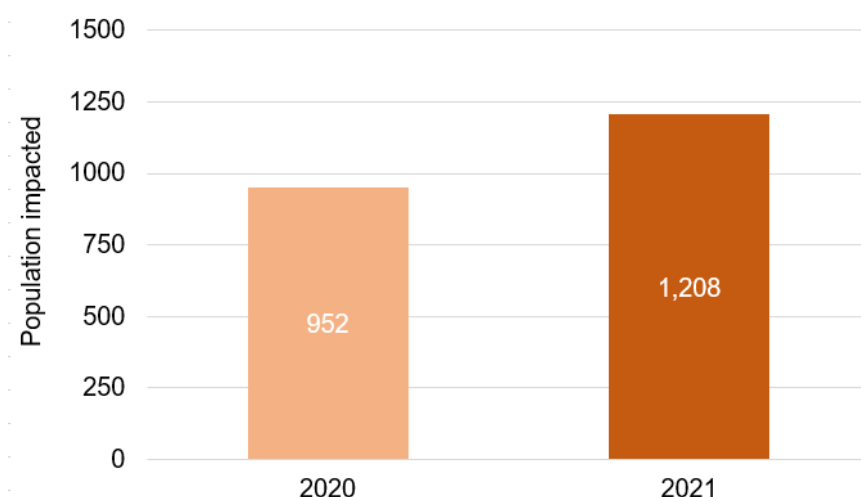


Figure 17 – Population impacted by long-term drinking water restriction notices³⁷

In its submission, Uisce Éireann outlined that the increase in population impacted by a boil water notice was largely driven by two water treatment plants: Lough Forbes in Longford (17,500 people) and Whitegate in Cork (9,897 people). While Lough Forbes was removed 33 days after being imposed, the CRU is concerned that the Whitegate supply was not removed for over five months, despite requiring two additional notices earlier in 2021. These two additional notices were also due to elevated turbidity (one short term and one long term).

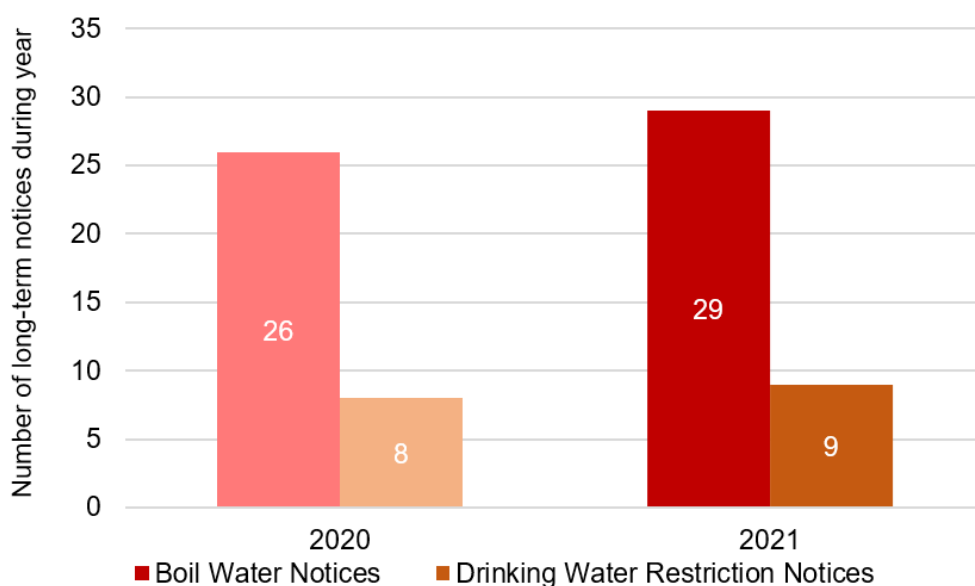


Figure 18 – Number of long-term notices in place³⁷

The CRU set Uisce Éireann a target that, by end 2021, 0 supplies should be impacted by a long-term boil water notice or drinking water restriction notice. This target expects Uisce Éireann to prioritise upgrades to assets to minimise the risk of supplies being put on a notice, and it also expects that in cases where notices are required to protect public health that Uisce Éireann responds swiftly to fix the issue. The target was delayed to 2021 – rather than 2020 – to allow time for Uisce Éireann to remove outstanding long-term boil water notices and drinking water restriction notices.

As evident by Uisce Éireann's performance, it did not meet this target by end 2021 as a number of long-term notices remained in effect after the end of 2021. This is despite removing critical outstanding notices by the end of 2020 (e.g., Lough Talt). In addition, as performance has declined since 2020, the CRU is concerned that Uisce Éireann is unlikely to meet this target within the RC3 period.

2.4 Sewer Incidents

2.4.1 Internal Sewer Incidents

Under this category the CRU will monitor the number of properties impacted by events of wastewater entering a building because a sewer is overloaded and the number of properties impacted by events of wastewater entering a building because of a failure of the sewer such as equipment failure, sewer collapse or a sewer blockage. The CRU will also monitor the number of properties at risk of having wastewater enter a building more frequently than once every ten years because of an overloaded sewer.

In the 2019 Implementation Update³⁵, the CRU stated that reporting capability under this metric has been in place since Q4 2019. Uisce Éireann at that time committed to reporting under this metric for the first time as part of the 2020 Annual Report, which the CRU included as a target in its decision on the Framework for 2020-2024.

Uisce Éireann did not meet this target to report on internal sewer incidents metrics. In its 2020 data submission (submitted to the CRU in 2021), Uisce Éireann stated it expected to first report to the CRU on internal sewer incidents in its next submission under the Framework i.e., this 2021 Annual Report. Uisce Éireann also committed to providing an estimate of the number of properties at risk of internal sewer flooding in this same submission.

Uisce Éireann has now failed to meet both the CRU set target as well as its own revised target to begin reporting on this metric. The CRU is concerned with the consistent delays in reporting on this metric despite confirming reporting capability as being in place since Q4 2019. Continued delays to reporting on this metric also impacts on the CRU's ability to set meaningful and appropriate targets to improve Uisce Éireann's performance, ultimately prolonging customer issues.

The CRU will continue to engage with Uisce Éireann to ensure reporting of this metric is fulfilled, including the establishment of baseline performance during the RC3 period so that the CRU can set appropriate targets.

2.4.2 External Sewer Incidents

The CRU's decision paper set out the inclusion of three new sub-metrics to monitor external sewer incidents. These sub-metrics will be aligned to that of internal sewer incidents through reporting under the same headings (overloaded, other causes, and properties at risk). This

³⁵ [CRU20140](#) Uisce Éireann Performance Assessment Framework 2019 Implementation Update

alignment also reflects how external sewer incidents are reported in other jurisdictions such as Northern Ireland and Scotland.

Further to discussions with Uisce Éireann during the consultation process, the CRU understands that Uisce Éireann is in a position to record both internal and external sewer incidents. Thus, the CRU considered it appropriate to include the external sewer incidents metrics at this time and set a target for Uisce Éireann to commence reporting with a view to establishing a baseline performance by end 2024.

In its 2021 submission, Uisce Éireann states that issues impacting on the reporting of internal sewer incidents are similarly impacting on the development of processes and systems to support reporting on external sewer incidents. The CRU notes that Uisce Éireann's submission, published alongside this report, does not provide an expected date by which it will report on these metrics. Uisce Éireann also did not provide an expected date in response to the CRU's follow-up questions regarding its submission. As with the reporting of internal sewer incidents, the CRU is concerned with delays to reporting on this metric. The CRU considers Uisce Éireann is not on track to report on external sewer incidents and establish a baseline performance by end 2024.

2.5 Environmental Performance

2.5.1 Incidents Relating to Wastewater

The CRU monitors two metrics regarding incidents relating to wastewater. The first looks at the number of pollution incidents resulting from wastewater collection and treatment activities, broken down by category, and the second looks at the number of recurring incidents.

The EPA classifies³⁶ an incident as:

- any discharge that does not comply with the requirements of a wastewater discharge licence, or;
- any occurrence at a wastewater works with the potential for environmental contamination or requiring an emergency response.

The EPA categorises incidents from 1 (minor) to 5 (catastrophic) depending on the potential impact to the receiving environment and/or human health. The CRU has targeted Uisce Éireann to have no Category 3, 4 or 5 pollution incidents during the RC3 period. In 2021, Uisce Éireann achieved this target.

1,067 once-off incidents occurred during 2021. While it is a slight reduction on the 1,108 incidents in 2020, it remains significantly above the number of once-off incidents recorded in the years prior to RC3 (see Figure 19 below). It also means an additional 482 once-off incidents above the CRU's set target of 585 or less occurred in 2021 and that Uisce Éireann missed its 2021 target by a greater margin than it did in 2020 when it had a target of 665 or less once-off incidents.

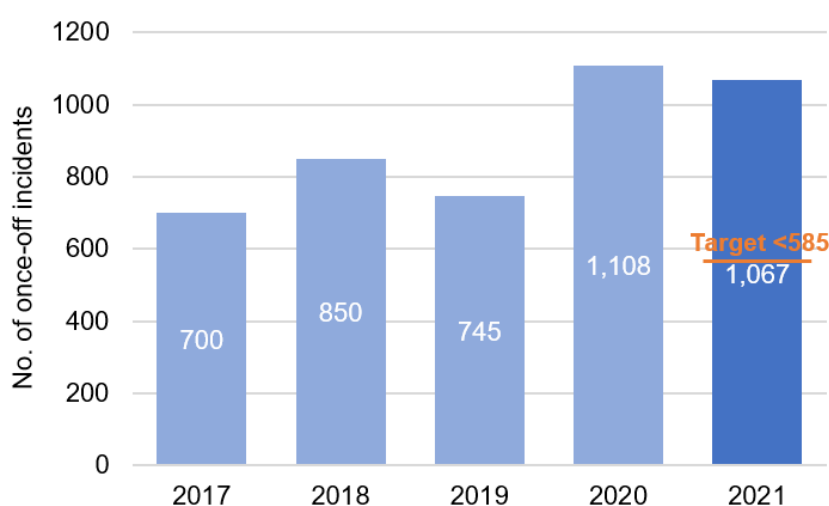


Figure 149 – Once-off Incidents

³⁶ [EPA Urban Waste Water Treatment in 2019](#), Appendix I: Environmental Incidents.

The CRU's target to reduce once-off incidents during the RC3 period includes a complete elimination of incidents that occur due to the operation, management and maintenance of wastewater treatment plants by the end of 2024. In 2021, 47% of once-off incidents were as a result of this.

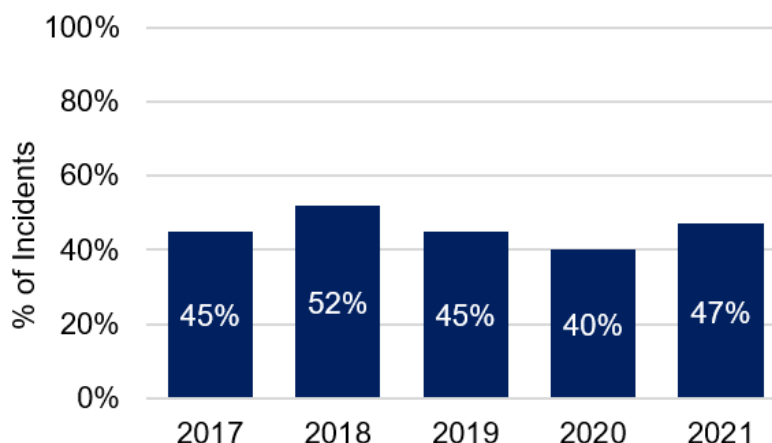


Figure 20 – Once-off Incidents Reported to EPA due to Operation, Management and Maintenance of Wastewater Treatment Plants

The EPA also monitors the number of recurring incidents open at the end of the year. These are incidents that are either ongoing or likely to recur until the underlying cause of the incident is resolved. At the end of 2021, 249 recurring incidents that remained open. Again, this does not meet the CRU's target under the performance assessment framework to reduce the number of recurring incidents to 170 or below in 2021. 17% of these recurring incidents were due to the operation, management and maintenance of wastewater treatment plants.

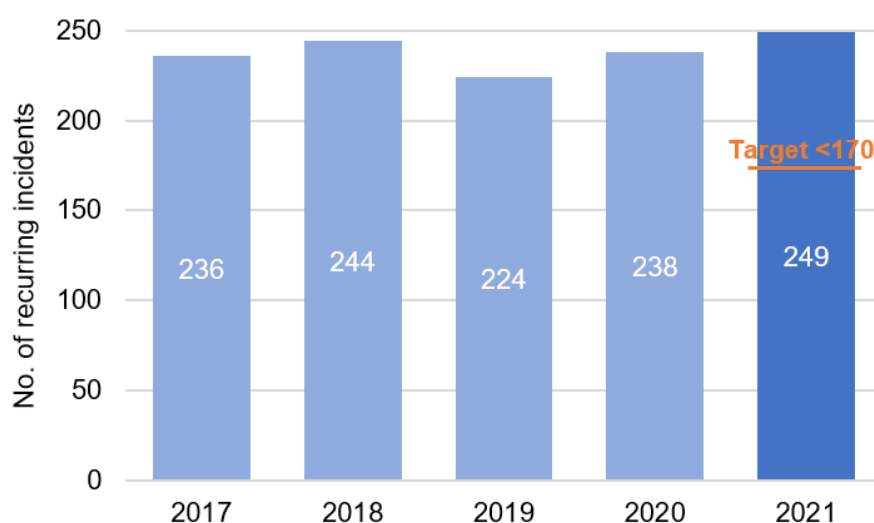


Figure 21 – EPA Recurring Incidents at Year End

The CRU welcomes Uisce Éireann's efforts to prevent the occurrence of Category 3, 4 or 5 pollution incidents in 2021. However, the CRU is concerned by Uisce Éireann's performance against all other targets under this metric.

The CRU is concerned with the significant gap between Uisce Éireann's annual performance and the CRU-set targets regarding the number of once-off and recurring incidents.

The CRU is also disappointed to note the increase of once-off incidents that occurred as a result of the operation, management and maintenance of wastewater treatment plants despite the CRU's explicit target to remove these incidents.

Finally, the CRU is concerned with the increasing trend of recurring incidents open at the end of the year since 2019 (the baseline year for RC3). Uisce Éireann remains not on track to meet the CRU's targets of <345 once-off incidents and <98 recurring incidents by the end of 2024.

2.5.2 Wastewater Agglomerations Meeting Treatment Requirements

The objective of the Urban Waste Water Treatment Directive (UWWTD) is to protect the environment from the adverse effects of urban waste water discharges.

A wastewater agglomeration is an area where the population or economic activities (or both) are sufficiently concentrated for a wastewater works to have been put in place. This metric consists of three sub-metrics: agglomerations with no wastewater treatment or preliminary treatment only, compliance with Emissions Limit Values (ELVs), and agglomerations compliant with the treatment and effluent quality standards of the UWWTD³⁷.

2.5.2.1 Agglomerations with no Wastewater Treatment

In 2013, there were 50 agglomerations in Ireland discharging untreated wastewater into the environment. Between 2014 and 2019, Uisce Éireann had completed work at a total of fifteen sites to reduce this number to 35.

For the RC3 period, the CRU approved spend for Uisce Éireann to provide treatment at the remaining 35 sites to bring this to zero. In 2021, Uisce Éireann was targeted to provide treatment at four sites. As no sites were targeted for 2020, this would result in 31 sites remaining on the list at the end of 2021.

Uisce Éireann completed work at one site in 2021, which, in addition to the one site provided treatment in 2020, leaves 33 sites discharging untreated wastewater into the environment at the end of the year. Uisce Éireann has subsequently missed its target for 2021.

³⁷ Council Directive 91/271/EEC

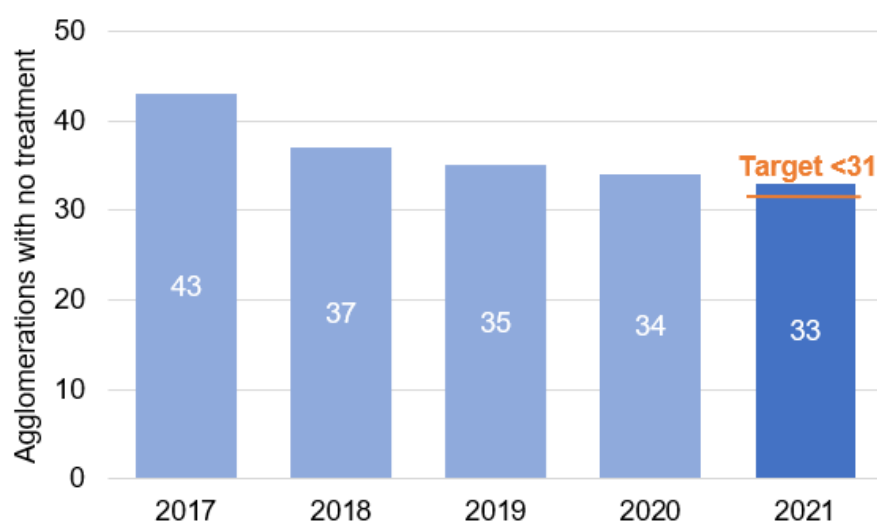


Figure 22 - Agglomerations with no Wastewater Treatment

The CRU is concerned that Uisce Éireann no longer plans to provide treatment at the remaining 33 sites before the end of 2024. At the time of writing this report, Uisce Éireann's webpage dedicated to eliminating raw sewage³⁸ (i.e., providing wastewater treatment for all agglomerations), stated that Uisce Éireann is on track to eliminate the majority of raw sewage discharges by the end of 2025.

This is an updated target set internally by Uisce Éireann and not in line with the CRU's RC3 targets. The CRU is concerned with the consistent delays in providing treatment at these agglomerations. This concern is compounded by the requirement to remove them under the Urban Waste Water Treatment Directive, a Directive which is currently being reviewed by the European Commission and as such is expected to include more stringent targets and treatment levels with Uisce Éireann will need to invest in to comply with.

As noted by the EPA in its Urban Waste Water Treatment Report 2021, the work to provide treatment for the agglomerations with no wastewater treatment, or preliminary treatment only, must be progressed and completed as soon as possible.

2.5.2.2 Compliance with the treatment requirements of Urban Waste Water Treatment Directive (UWWTD)

The UWWTD sets requirements for the collection and treatment of wastewater from large urban areas to protect the environment. It also sets quality limits that the treated wastewater must meet

³⁸ [Uisce Éireann](#) - Eliminating raw sewage

depending on the size of the urban area and the type of water body the treated wastewater is discharged to.

The CRU understands from Uisce Éireann that the number of agglomerations subject to the UWWTD varies on an annual basis.

There were 174 agglomerations subject to the UWWTD in 2021, the same number of agglomerations subject to the UWWTD in 2020. As with 2020, 162 were compliant with the treatment and effluent quality standards of the Directive and 12 agglomerations were non-compliant.

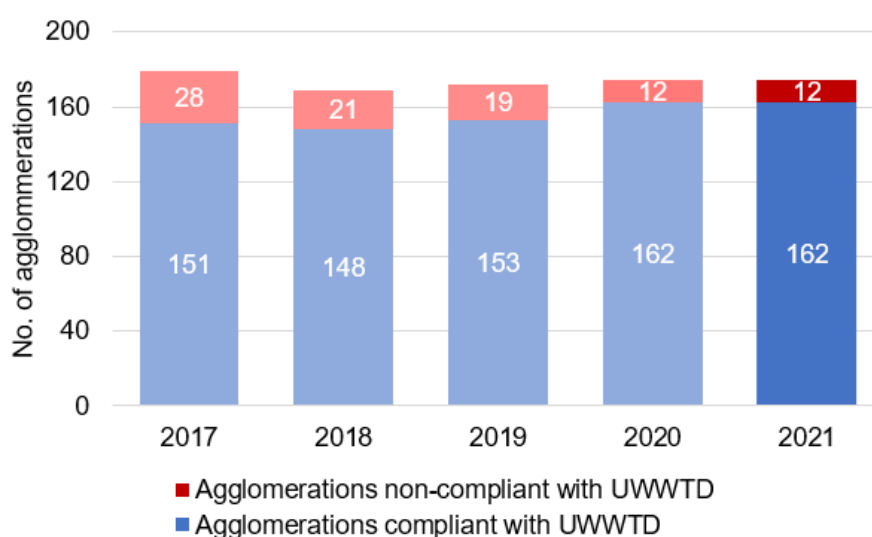


Figure 253 - Agglomerations subject to the UWWTD

However, these 12 non-compliant agglomerations represent 49% of the agglomeration population equivalent that fall under the UWWTD. This means that urban areas comprising almost half of the population covered by the UWWTD were not compliant with the standards in the Directive. This is largely attributed to the non-compliance of the Ringsend Wastewater Treatment Plant in Dublin, which treats almost half of Ireland's urban wastewater.

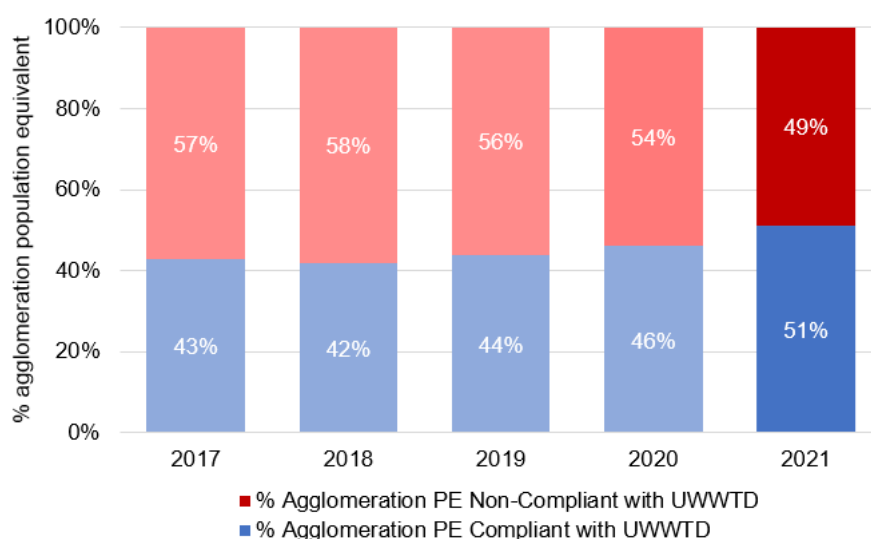


Figure 24 - Percentage Agglomeration Population Equivalent subject to UWWTD

In its decision on the Framework for 2020-2024, the CRU set Uisce Éireann a target to be 100% compliant with the treatment requirements of the UWWTD by the end of the RC3 period. Uisce Éireann explained that the population equivalent compliance with the UWWTD is an improvement on previous years is due to the increased collected load within the compliant agglomerations, in line with population growth, which resulted in an overall increase in the percentage agglomeration population equivalent compliant. However, it remains far below the CRU's target of 100% by end 2024.

Uisce Éireann is not currently on track to meet this target as Uisce Éireann has revised the timeline to complete the upgrade and expansion of Ringsend to 2025. Delays in providing these sites with adequate treatment are of particular concern in light of the Court of Justice of the European Union declaring that Ireland has failed to fulfil its obligations under the Urban Waste Water Treatment Directive in 2019.

2.5.2.3 Compliance with the Emission Limit Values for Urban Wastewater Licences

As mentioned earlier, all of Uisce Éireann's wastewater discharges to the environment requires a wastewater discharge licence or certificate of authorisation from the EPA. Wastewater licences set out conditions relating to the licensed wastewater discharges to ensure the treated effluent does not negatively impact on the environment.

The licensing of wastewater treatment plants provides limits on wastewater discharges to protect the receiving water environment where the discharge occurs. Achieving compliance with these limits is essential to protecting our water quality. This metric looks at specific parameters to provide an overview of Uisce Éireann's performance.

This is the second time Uisce Éireann has reported to the CRU on its compliance with ELVs. Further information on previous performance is available on the EPA's website.³⁹

In 2021, 43% of Uisce Éireann's wastewater treatment plants met the overall compliance with ELVs. Uisce Éireann's highest compliance rate with a single parameter was with the COD limit, of which 86% of wastewater treatment plants complied with. The lowest compliance rate with a single parameter was in relation to the Ammonia limit, where 60% of wastewater treatment plants were compliant with the limit. Further information is provided in Figure 25 below.

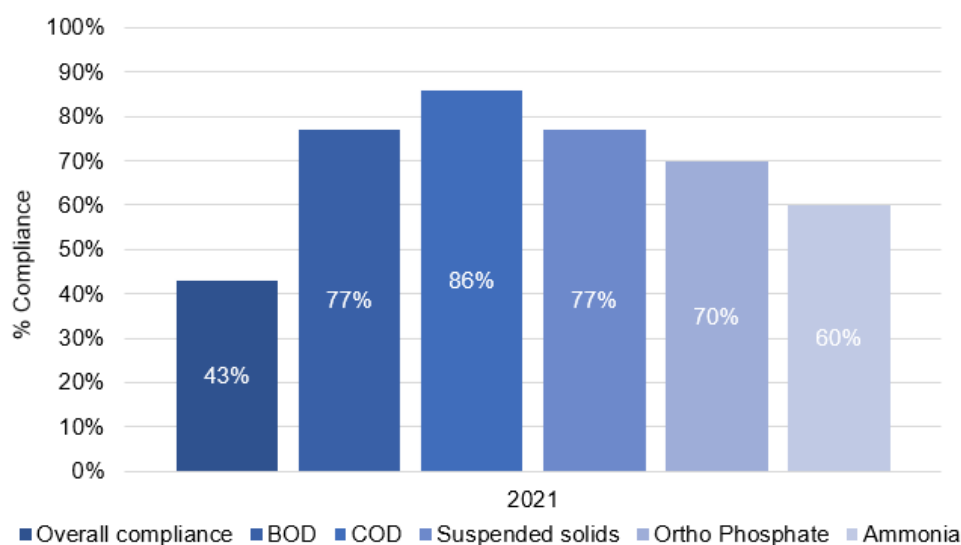


Figure 25 - Compliance with Emissions Limit Values in 2021⁴⁰

In its decision paper, the CRU set out that Uisce Éireann is to target establishing a baseline of performance by 2024. It remains on track to establish a baseline performance. The CRU will continue to monitor Uisce Éireann's performance here during the period and will set targets for RC4 once a baseline has been established.

³⁹ The EPA [Urban waste water licence compliance](#)

⁴⁰ Note the CRU uses Uisce Éireann's submitted figures at present as it establishes a baseline performance. These may differ to figures published by the EPA. The CRU will consider this when setting targets under this metric.

2.5.3 Sludge Reuse and Disposal

This metric will measure the percentage of drinking water and wastewater sludge that is disposed of in a satisfactory manner.

Uisce Éireann has reported on drinking water sludge for the first time in 2019. Uisce Éireann disposed of 100% of its drinking water sludge in a satisfactory manner in 2021.

In 2021, 8,752 tonnes of drinking water sludge (in total dry solids) were produced, a significant increase on the 5045.82 tonnes produced in 2020. Uisce Éireann attributes the increase to new water treatment plants coming into operation. 91% of this was sent to sustainable outlets. Sustainable outlets refer to ways in which the sludge can be recovered and reused rather than sent directly to landfill for disposal. Drinking water sludge is typically used in manufacturing industries as a sustainable outlet, such as in cement manufacturing where aluminium sludge can displace bauxite as a raw material in cement production. The remaining 9% was sent to landfill capping.

A breakdown of the percentage total dry solids sent to sustainable outlets and landfill capping since 2019 is provided in Figure 26 below. The associated amounts in tonnes total dry solids (TDS) is also provided.

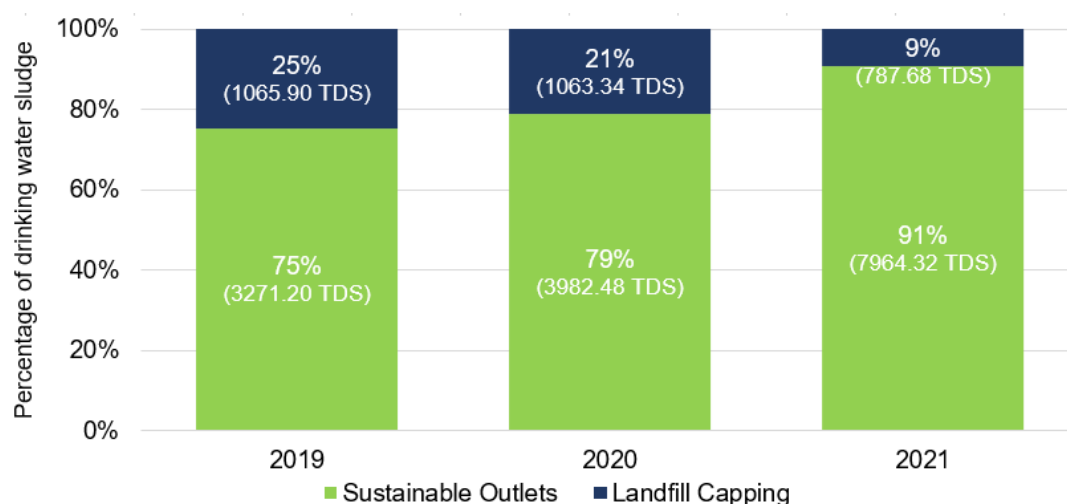


Figure 26 - Drinking Water Sludge

The CRU greatly welcomes the increase in the percentage of drinking water sludge sent to sustainable outlets, particularly at a time of such significant increase in the amount of drinking water sludge produced. Such improvements are especially valuable in promoting a circular economy.

In its decision on the Framework, the CRU set Uisce Éireann a target to begin reporting on the wastewater sludge metric in 2023 (with full-year 2022 data). In its place, until this metric is reported on in full, Uisce Éireann is to provide the tonnes of wastewater sludge produced annually and final disposal/recovery outlets to which it is sent, in line with how Uisce Éireann reports on this to the EPA.

In 2021, 60,467 tonnes dry solids of wastewater sludge were produced. The final disposal/recovery outlets and related amounts are outlined in Table 2 below. Further information on wastewater sludge is available in the EPA's Urban Waste Water Treatment in 2021 report⁴¹.

Table 2 – Wastewater Sludge Produced in 2021

Total Wastewater Sludge (tonnes dry solids)	Agriculture	Compost	Landfill	Other
60,467	55,182	5,154	0	131

⁴¹ The EPA [Urban Waste Water Treatment in 2021, Appendix H: Sewage sludge](#)

2.6 Energy and Emissions

2.6.1 Energy Consumption

In its decision on the Framework for 2020-2024, the CRU included a new metric to monitor Uisce Éireann's Total Primary Energy Requirement (TPER) in GWh. As a public sector body, Uisce Éireann must report on its performance to the Sustainable Energy Authority of Ireland (SEAI) annually, with performance targets set by the government in the National Energy Efficiency Action Plan and Climate Action Plan. The CRU set Uisce Éireann a target of a 40.71 GWh in Total Primary Energy Requirement (TPER) minimum reduction by end 2024.

Uisce Éireann's submission under this report does not include energy consumption figures for 2021 as the data has not yet been verified by the SEAI. However, it does confirm the data and Uisce Éireann's performance against public sector targets for 2020, which have been verified by the SEAI.

In 2020, Uisce Éireann's energy consumption as reported by the SEAI was 1,059.6 GWh in TPER, exceeding its obligation to achieve 33% energy efficiency by 2020⁴² by achieving 34.5% energy efficiency. The CRU welcomes Uisce Éireann's commitment to meeting its public sector targets, particularly as it continued to be the highest single public user of energy in 2020.

2.6.2 Greenhouse Gas Emissions

Along with energy consumption, the CRU also included a new metric to monitor Uisce Éireann's energy-related emissions in CO₂ equivalent in line with its reporting to the SEAI.

The CRU understands that 2022 is the first year public sector bodies will be reporting under the new greenhouse gas emissions target for 2021. As with energy consumption, 2021 data has not yet been verified by the SEAI.

In the interim to the SEAI reporting on this for the first time, the CRU will continue to provide Irish Water's emissions in CO₂ equivalent that are produced by Irish Water's energy consumption, reported above. Uisce Éireann's submission under this annual report confirms the data provided for 2020 that has since been verified. In 2020, Uisce Éireann's primary energy consumption equated to 166,138,123 kgCO₂ of energy-related CO₂ emissions.⁴³ This is a reduction of 20,477,536 kgCO₂ on 2021. Further information will be provided in the next report.

⁴² Department of Communications, Energy and National Resources [The National Energy Efficiency Action Plan 2009 – 2020](#)

⁴³ SEAI [Annual Energy Statement 2020 for Uisce Éireann](#)

3. CRU Assessment

3.1 Performance Assessment

An overview of the CRU's assessment of Uisce Éireann's progress to 2024 targets for each metric is summarised in Table 1. The CRU will continue to monitor Uisce Éireann's progression towards 2024 targets to the end of the period.

The CRU set Uisce Éireann targets for 2021 on 12 of the 27 metrics in the decision paper. In terms of progression to 2024, Uisce Éireann's 2021 submission provided full performance data on 15 of the 27 metrics set out in the CRU's decision paper, an increase of one compared to 2020 reporting.

Of the 12 metrics with targets for 2021, a total of 19 targets were set against these (some metrics had more than one target). Uisce Éireann has met nine of these 19 targets, resulting in Uisce Éireann missing ten targets. In 2020, Uisce Éireann met eight of 21 targets set for that year.

It is worth noting that the CRU's targets, while fair and appropriate, were decided and published in August 2021. Therefore, Uisce Éireann would not have been in a position to put in place all improvements for the years of 2020 and 2021. Notwithstanding this, the CRU is still concerned with some of the declining performance and trends identified from the years 2019 to 2021. The CRU expects improved performance in the latter years of RC3.

Overall, the CRU is concerned with the level of underachievement projected not to meet targets within the RC3 period. The CRU has requested Uisce Éireann to provide an update on measures it will take to meet the agreed targets for those investments by 2024. In parallel, the CRU will review the findings of the review of the implementation of the Scottish Water International recommendations and assess the impact on the Performance assessment framework and capital investment plan delivery.

In addition, the CRU requests sight of Uisce Éireann's new quality assurance process for the Investment Plan approval, prior to submission to the CRU in June 2023.

Subject to accurate and timely submission of by Uisce Éireann of its 2022 report, the CRU will publish its assessment of these reports.

4. Next Steps

The CRU will continue to engage with Uisce Éireann to implement the remaining metrics included in the CRU's decision on the Framework during the RC3 period. This includes reviewing Uisce Éireann's leakage data, implementation of the security of supply metric and revision of the customer satisfaction survey, among others. A status update regarding the metrics remaining to be implemented is provided in the Table below.

Metric	Steps Required to Implement	Status
Customer Satisfaction Survey	UÉ to implement changes to survey from January 2024	Pending implementation (delayed from 2023 to 2024)
Stakeholder Engagement Panel	CRU to establish. CRU will consult on Terms of Reference for the Panel in 2023.	In progress
Security of Supply Metric	UÉ to report on security of supply	UÉ to report (delayed)
Customer Supply Pipe Leakage	CRU to engage with UÉ on submitted data	In progress
Internal Sewer Incidents	UÉ to report, CRU to engage regarding 'at risk' and set targets	UÉ to report (delayed)
External Sewer Incidents	UÉ to report, CRU to engage regarding scope of metric, 'at risk', and set targets	UÉ to report with baseline by end 2024
Greenhouse Gas Emissions	CRU to consider if further targets appropriate as scope on UÉ's public sector obligations has been set	In progress

Uisce Éireann's next submission regarding 2022 performance is due to be submitted in Q3 2023. The CRU will assess this submission and publish its assessment of Uisce Éireann's performance here in Q4 2023.

Appendix A

Real losses on Uisce Éireann's network, commonly referred to as leakage, includes leaks on trunk mains and distribution pipes, leaks on service connections and leaks and overflows at storage reservoirs. There are two approaches to determining leakage on the public network. The first looks at a top-down water balance where the water entering the network is assigned to water losses and water use based on metering information and well-reasoned estimates (see Figure 27 below).

Distribution Input	Authorised Use	Non-Domestic Customers	Non-Domestic Use
			Internal Plumbing Losses
			Supply Pipe Leakage
		Domestic Customers	Domestic Use
			Internal Plumbing Losses
			Supply Pipe Leakage
	Unbilled Water		Irish Water Use
			Other Authorised Unbilled Use
	Water Losses	Apparent Losses	Unrecorded Connections
			Metering and Data Errors
		Real Losses on Irish Water's Network	Leaks on Service Connections
			Leaks on Trunk and Distribution Mains
			Leaks & Overflows at Storage Reservoirs

Figure 27 - Components of Water Balance

In addition to this, water losses should also be estimated using a bottom-up approach by monitoring demand at a time when customer use is low (typically overnight). During a period of low predictable customer use, flow into District Metered Areas (DMAs) is monitored for a continuous period of at least one hour. This flow is then allocated between public network losses, customer supply pipe losses and customer use and then converted from an hourly to daily rate

with an adjustment made for variations in pressure between day and night. Estimates of losses on trunk mains and service reservoirs are then added to the calculated DMA losses to provide an estimate of total losses on the public network.

A final leakage number can then be reported by reconciling differences in the top-down and bottom-up approach to leakage estimation and applying robust statistical analysis in line with best international practice.

The CRU has also requested that Uisce Éireann provides an estimate of how much of the water delivered to customer properties is lost to leaks on the private side. Leaks on the private side can occur on the customer supply pipe and in internal plumbing.

Leaks on domestic customer's external customer supply pipe (see Figure 28 below) may qualify for a free repair by Uisce Éireann under the First Fix Free Scheme. Under this scheme, Uisce Éireann notifies metered domestic customers when it suspects a leak is occurring within the boundary of their property. A leak alarm notifies Uisce Éireann that there is a constant flow of six or more litres of water per hour for a continuous period of 48 hours or more. Savings from the scheme results from a mix of repairs on the customer supply pipe carried out by Uisce Éireann and repairs carried out by customers after being informed by Uisce Éireann that the leak alarm on their meter had been triggered.

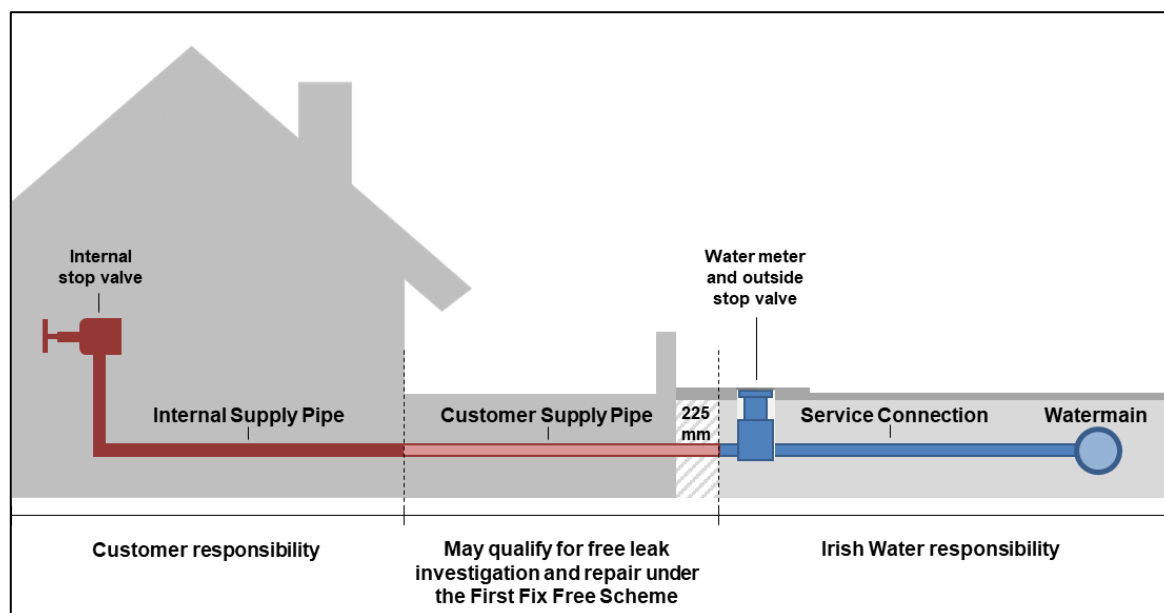


Figure 28 - Pipe Responsibility First Fix Free Scheme